Windows Deployment Service

What is Imaging?

Imaging is the process of creating a complete backup (or image) of a computer's system, including its operating system, installed software, settings, and data. The image is typically saved as a single file and can be used to quickly replicate the same system setup on other computers. It involves capturing a snapshot of the computer's current state, which can be deployed to other machines.

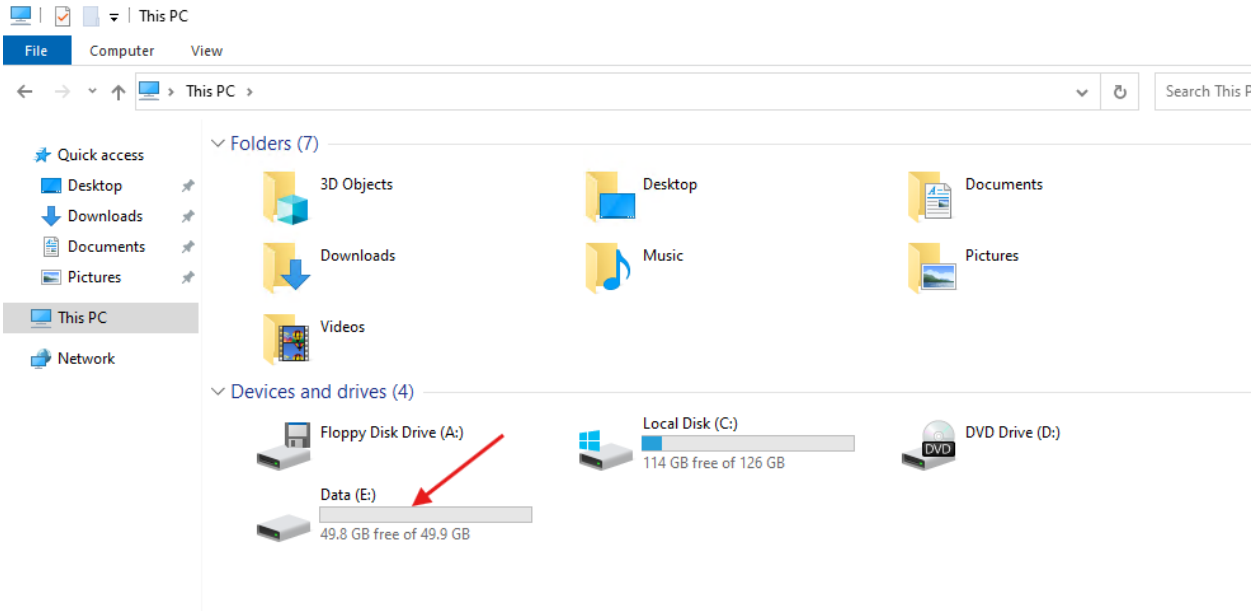
Why Does Imaging Matter for Organizations?

Imaging ensures standardization by making sure all systems across the organization have the same software, configurations, and security settings. It saves time by speeding up the deployment of new machines or the reinstallation of old ones, removing the need for manual setup. Imaging promotes consistency, helps avoid discrepancies in software versions or configurations, and simplifies IT management by making it easier to maintain and update systems. It also supports disaster recovery, allowing quick restoration of systems in case of failures, and enhances security by ensuring all systems have the same security configurations, reducing vulnerabilities.

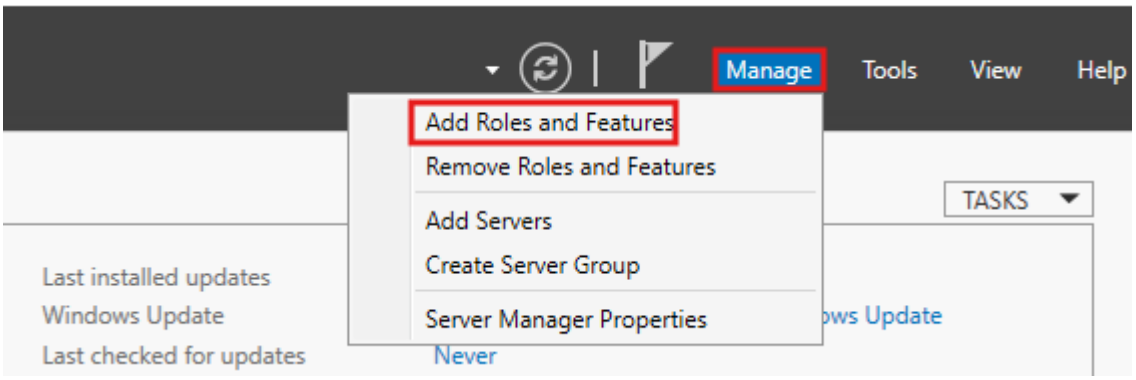
Imaging is essential for organizations to save time, maintain uniformity, and streamline IT operations.

To perform this task, I have configured a Domain Controller named 'DC,' and the domain name is 'rangelitech.online.' Let's start the installation of the WDS service on this server.

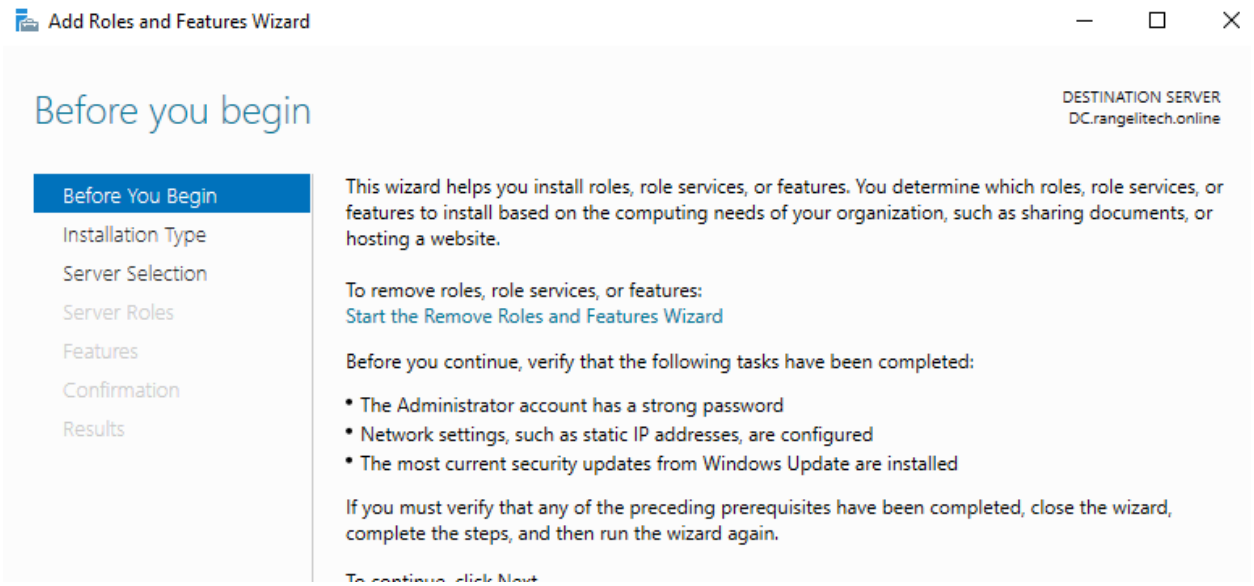
Before continuing, add a hard drive to the Domain Controller, which you will need later to save the Remote Installation Folder.



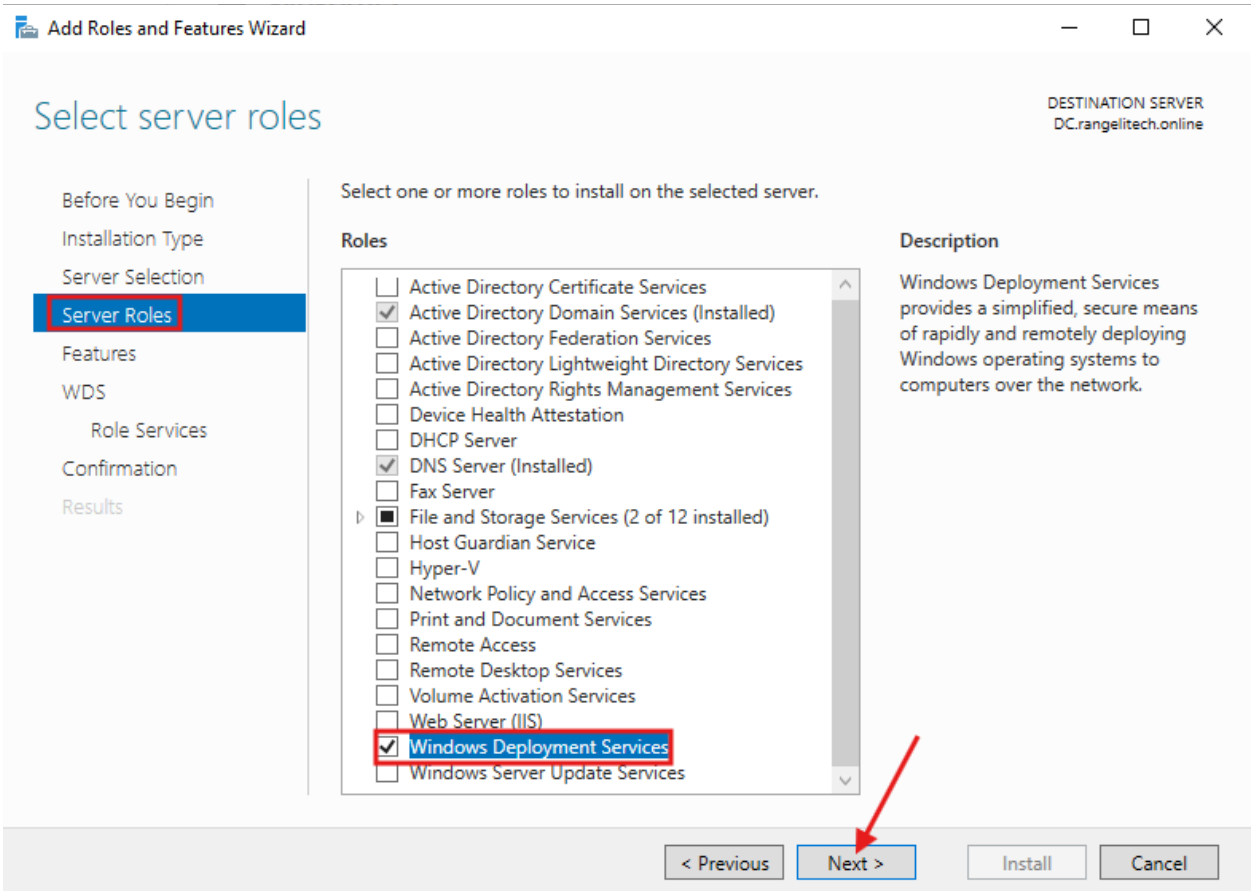
Go to Manage and click on Add Roles and Features.



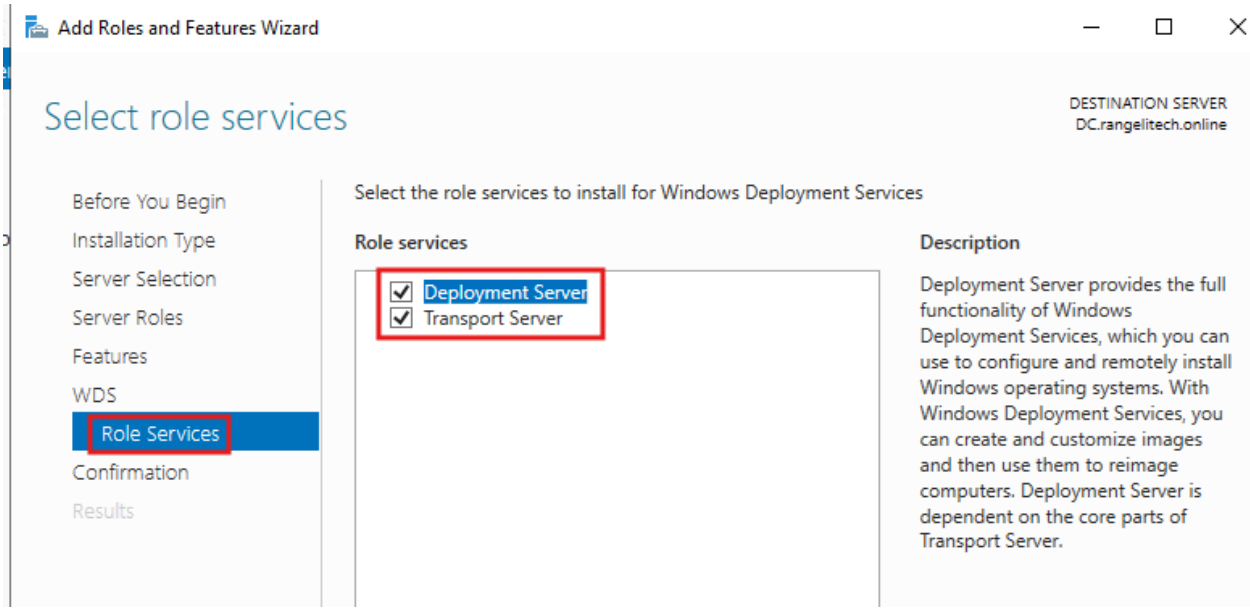
Stat the Installation.



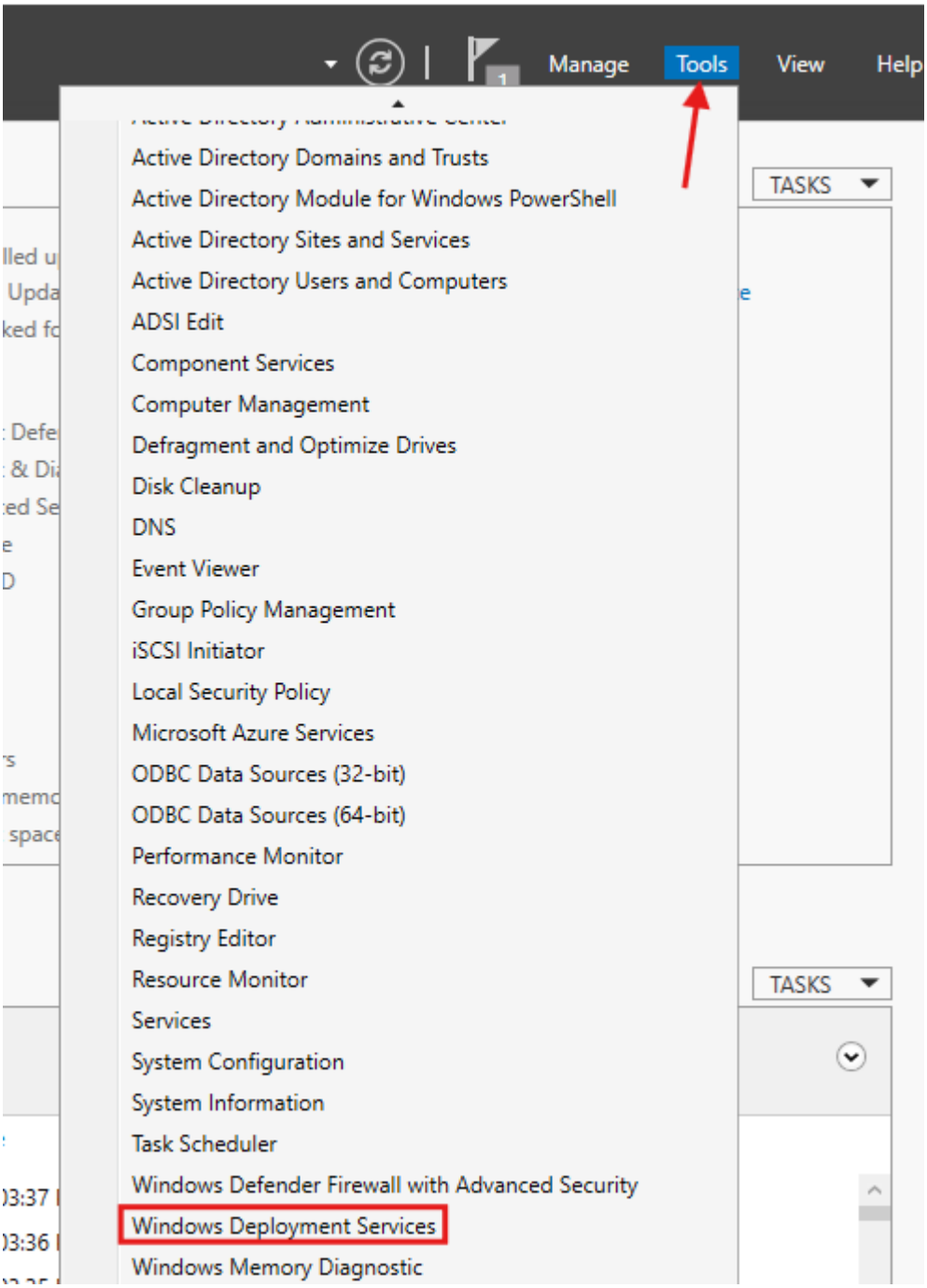
On Server Roles, check Windows Deployment Services and click Next.



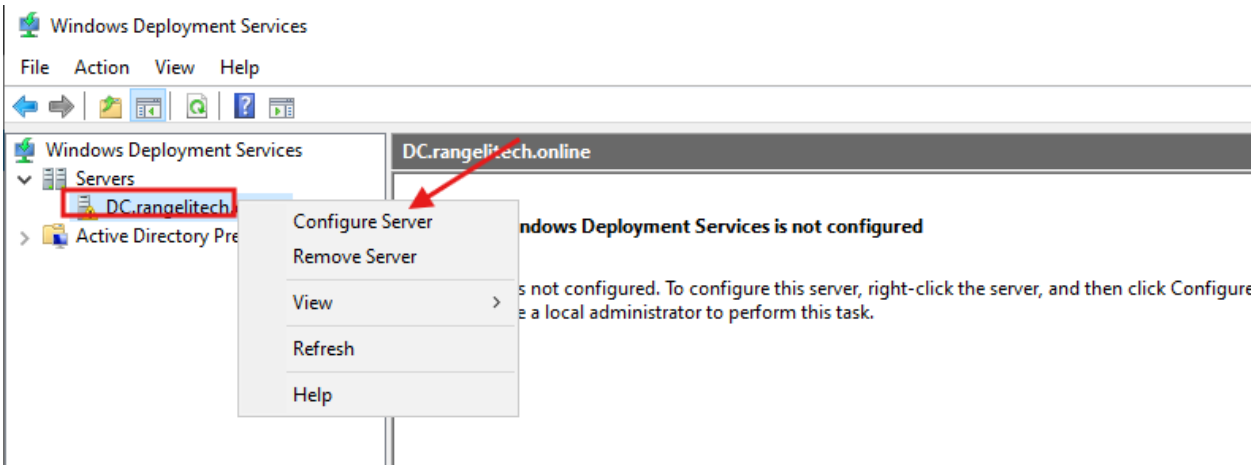
Here, you need to select both services.



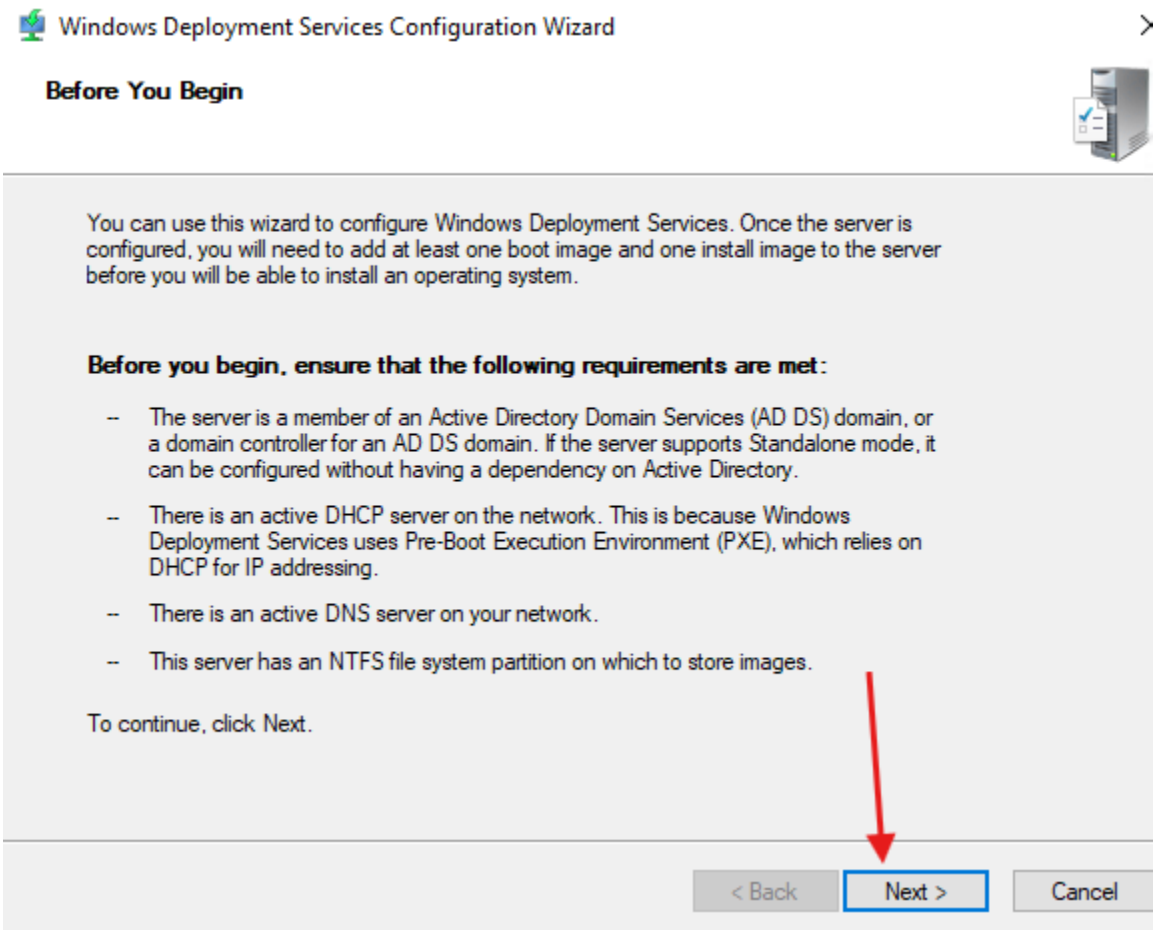
After installing the WDS server role, we need to configure the WDS service. To do this, open the WDS Management Console.



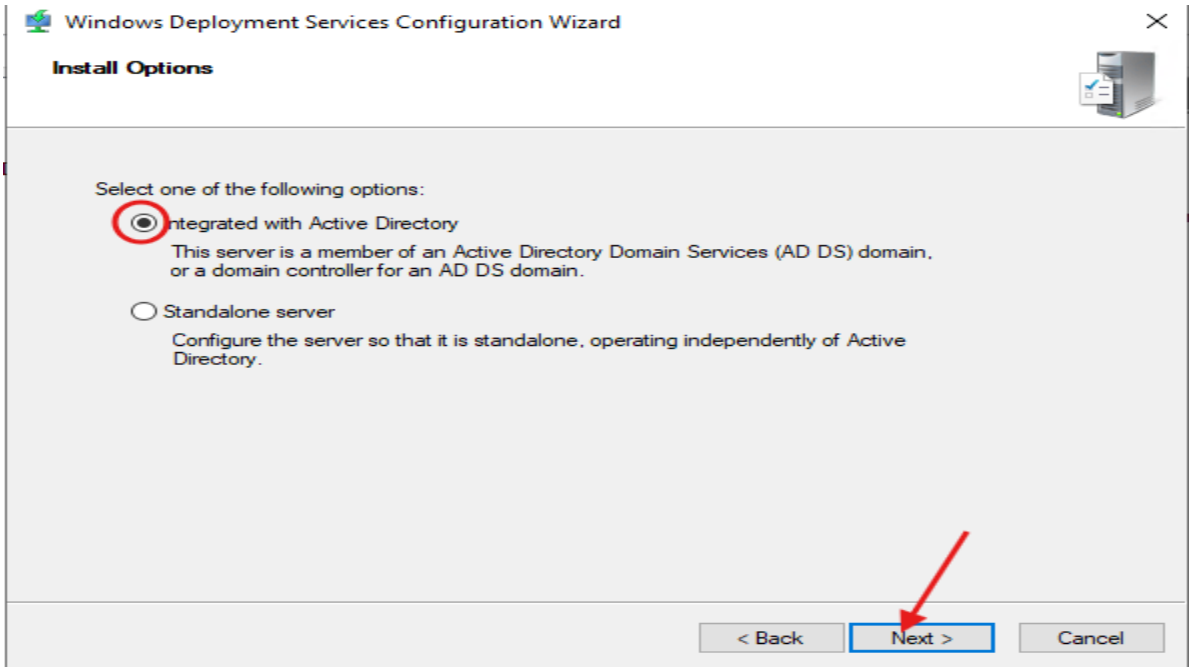
Right-click on the server and click Configure Server.



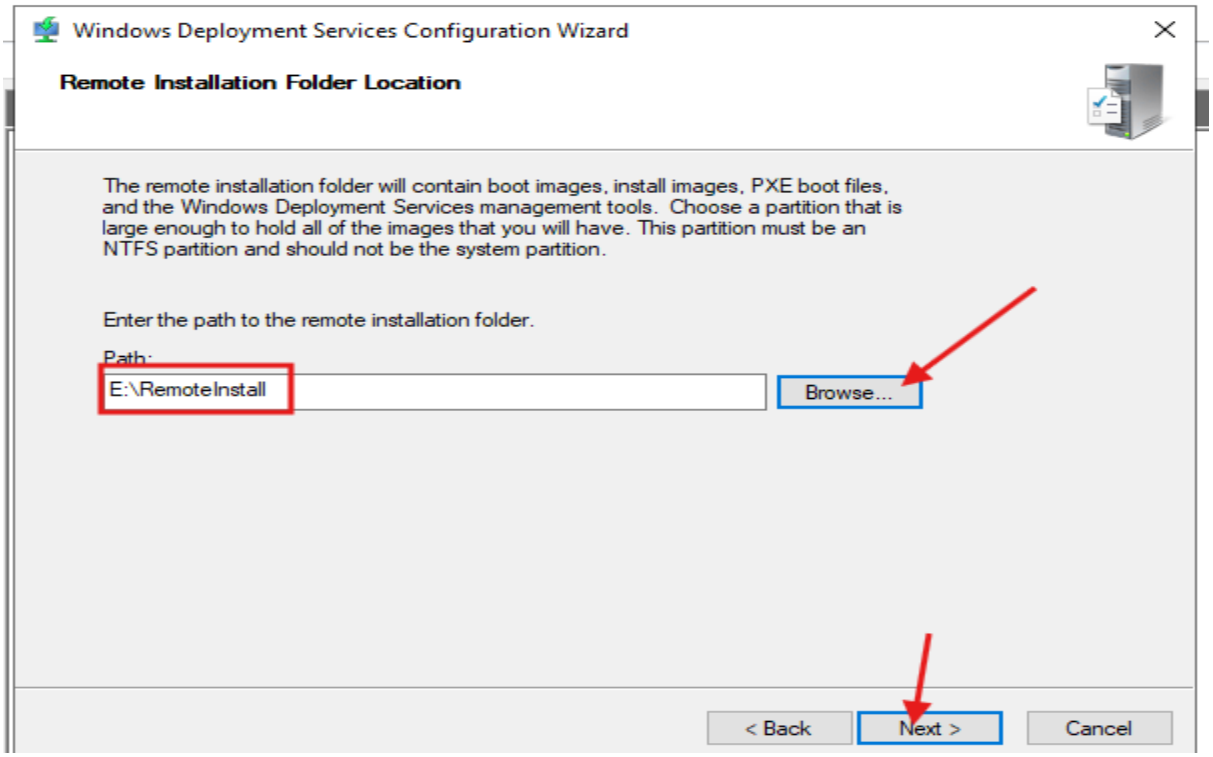
In this wizard, you can see the requirements for configuring the WDS service.



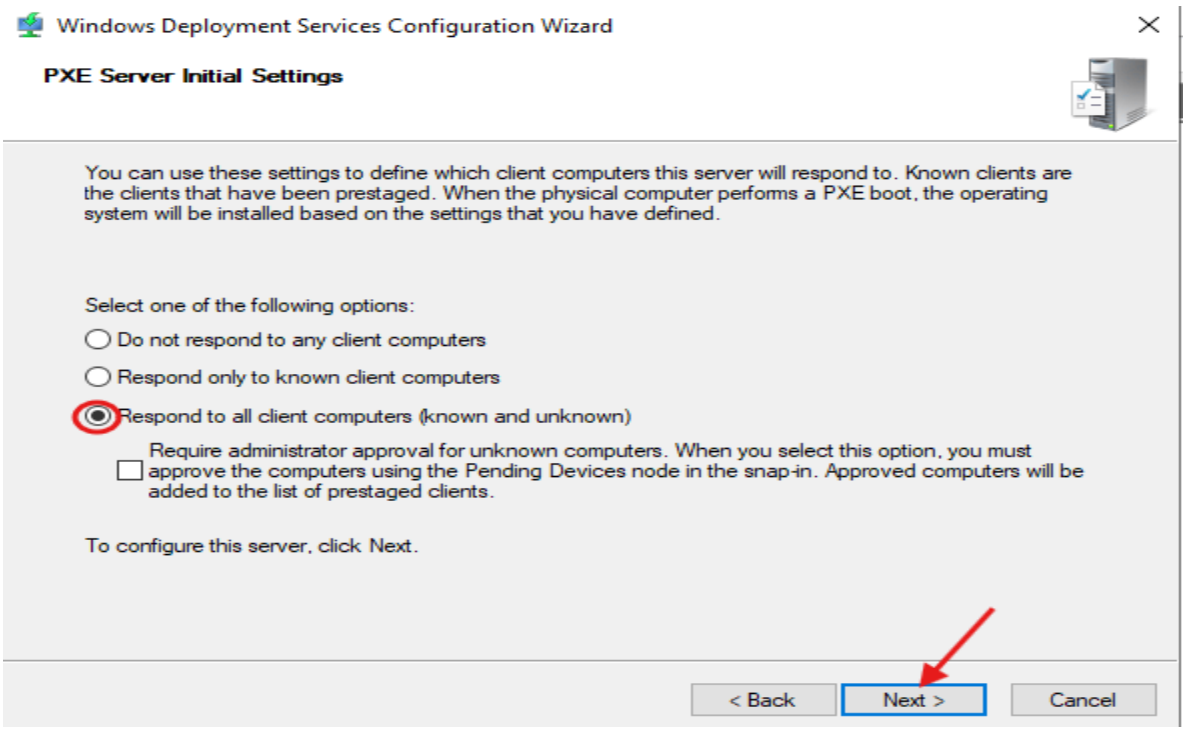
Here, you will go with the default selection.



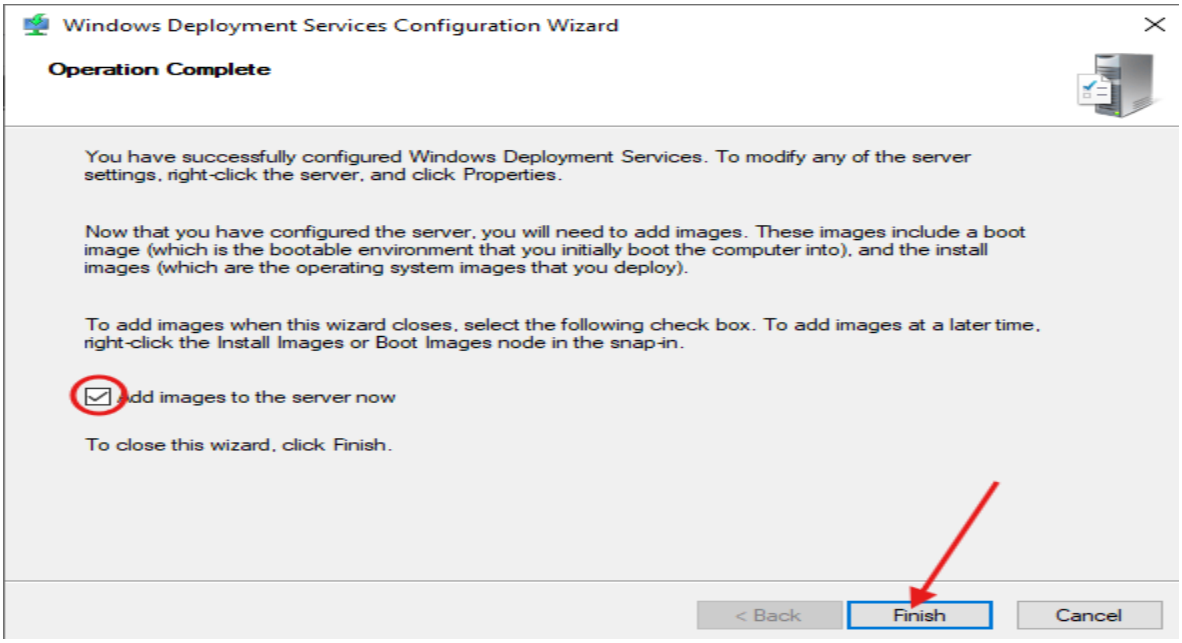
Here, it will ask you for the location of the RemoteInstall folder. You can specify the path of the drive you added when configuring the Domain Controller.



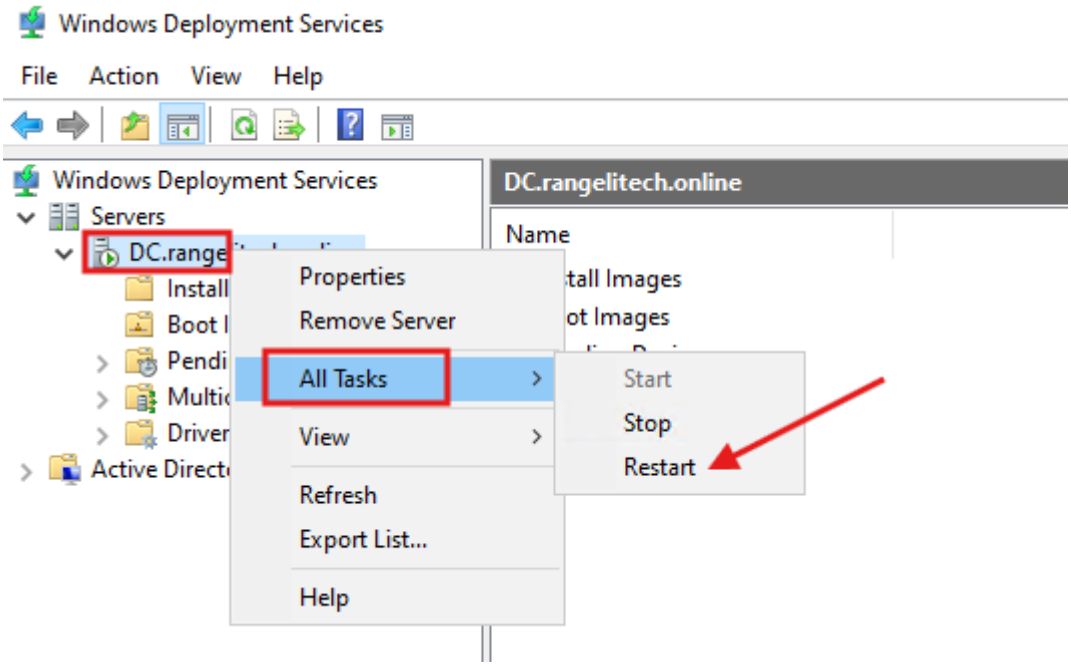
Here, you will select Respond to all client computers (known and unknown). If you want, you can check the box below as per requirements but for this testing purpose, you can leave it unchecked.



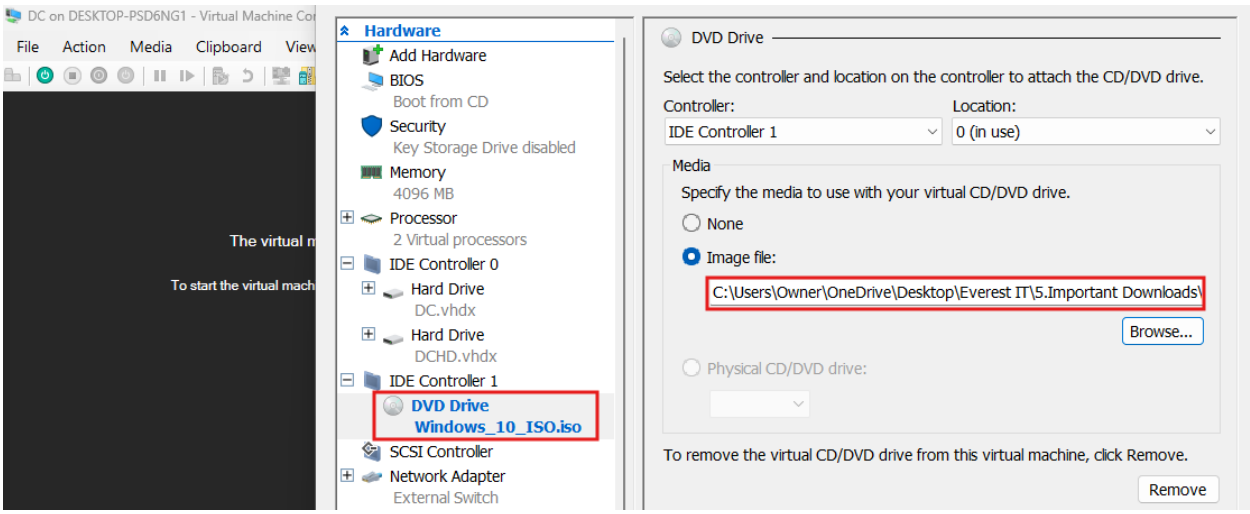
Once the configuration is completed, click the Finish button.



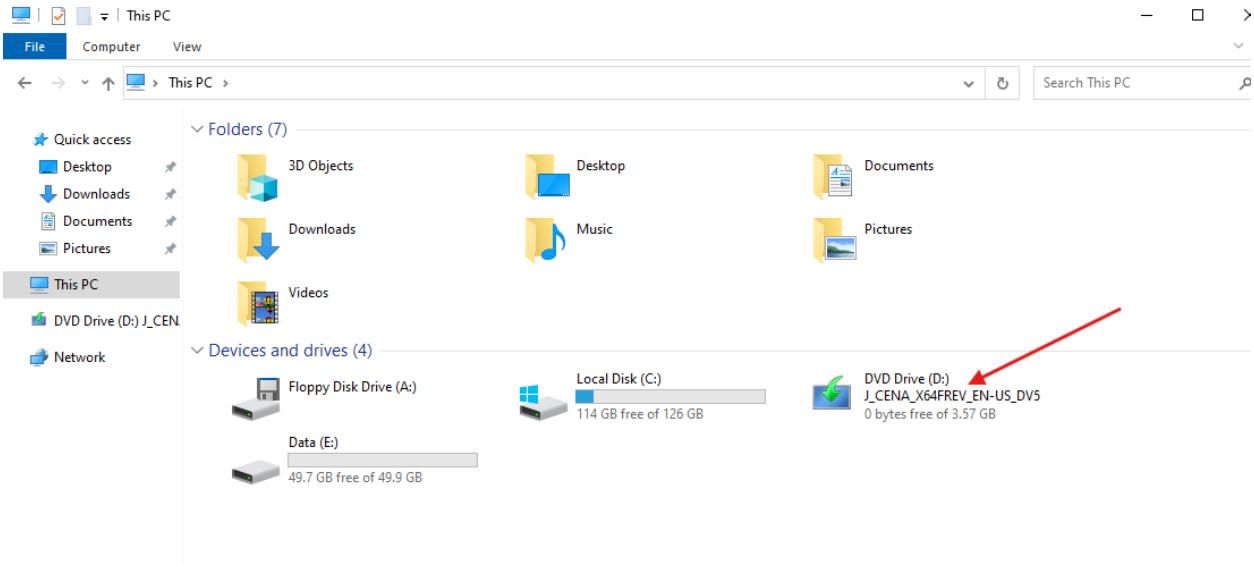
After the configuration setup, you can restart the service.



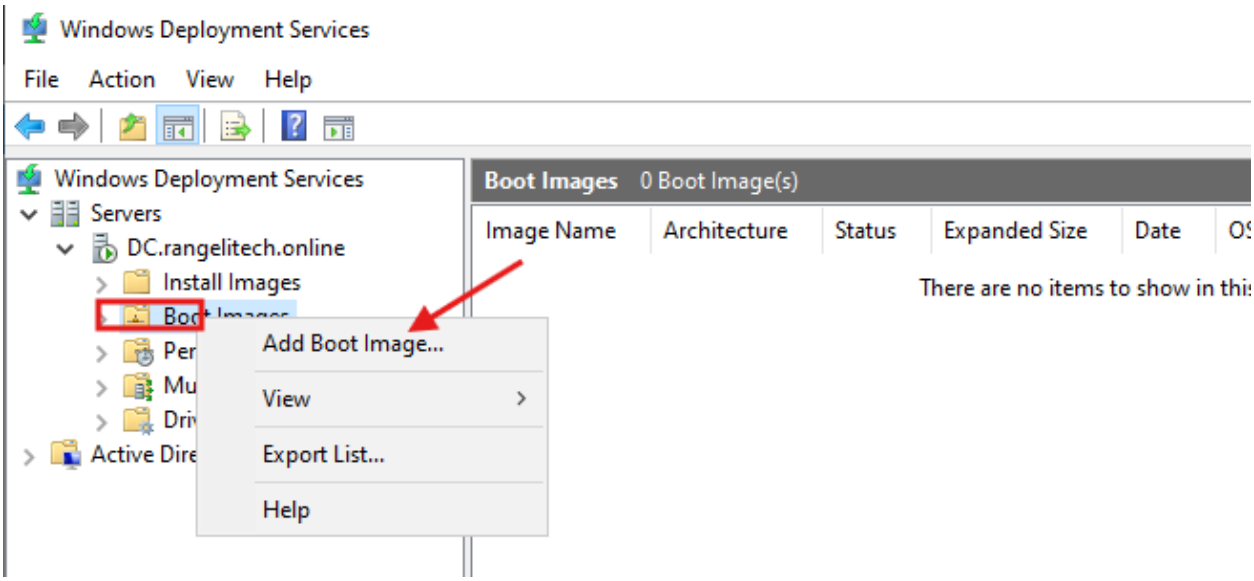
Now attach the operating system to the server that you want to deploy. For this task, I have attached the Windows 10 operating system.



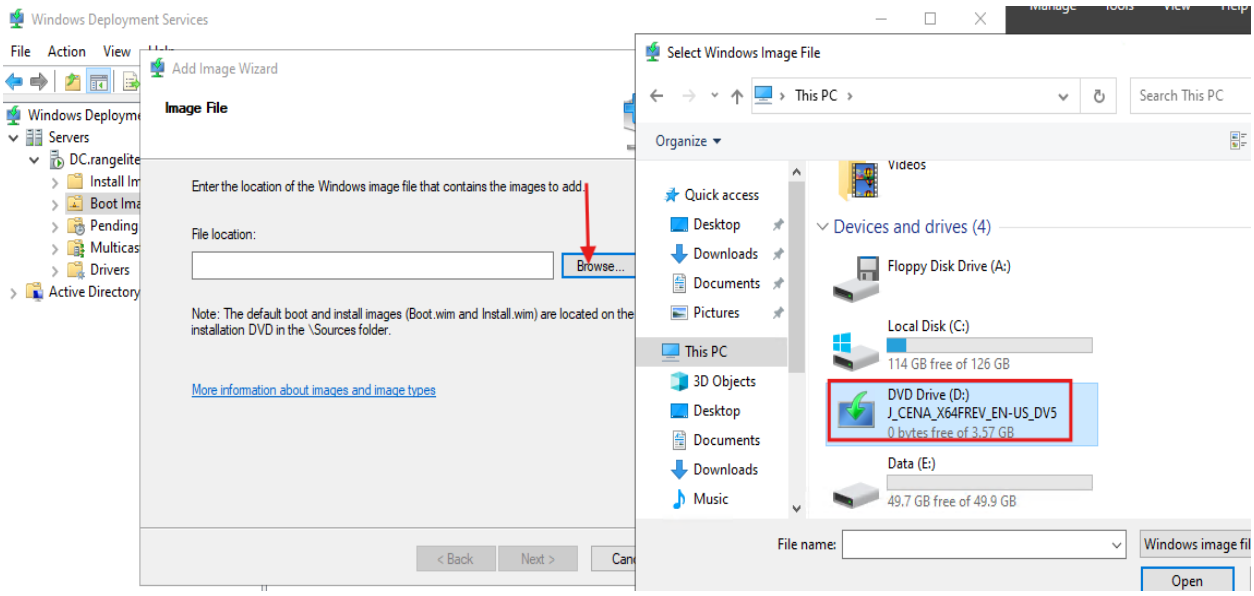
Here, you can see that the operating system ISO file in the D drive has been successfully attached.



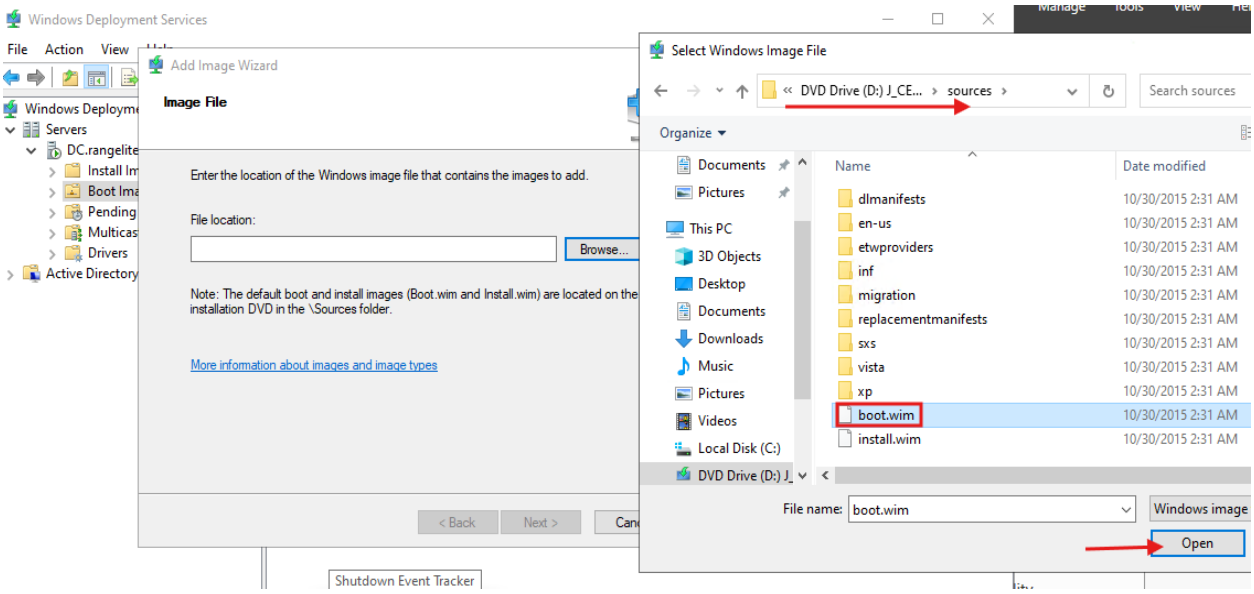
Let's go back to the WDS Management Console to add a boot image. Select Boot Images, right-click on it, and select Add Boot Image.



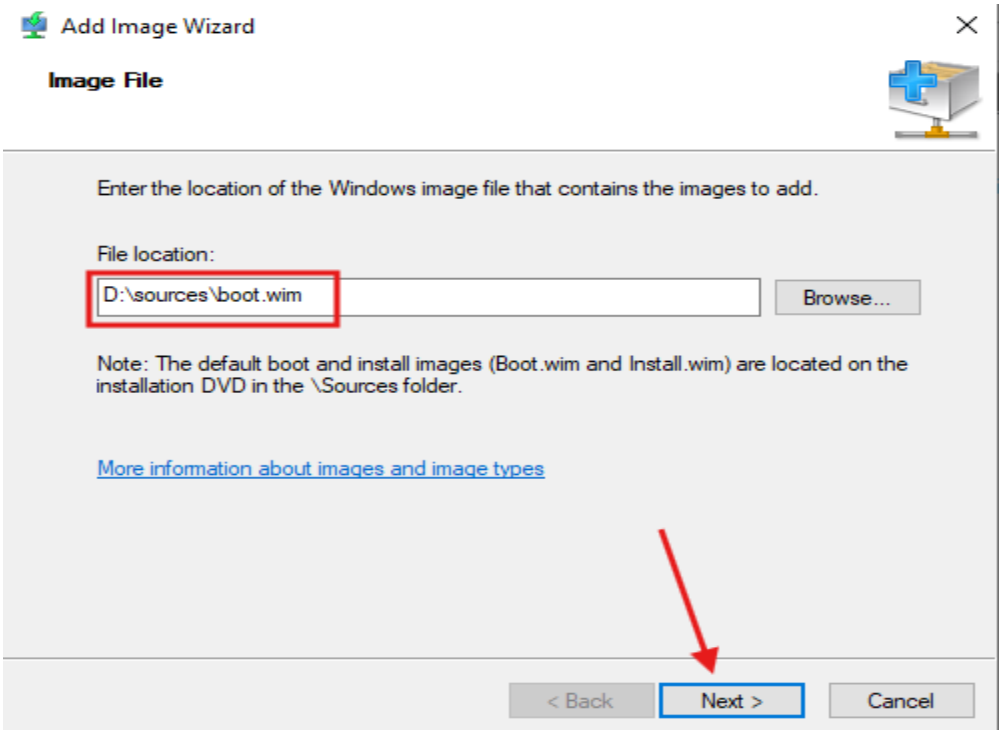
It will ask you to specify the location of the boot image.



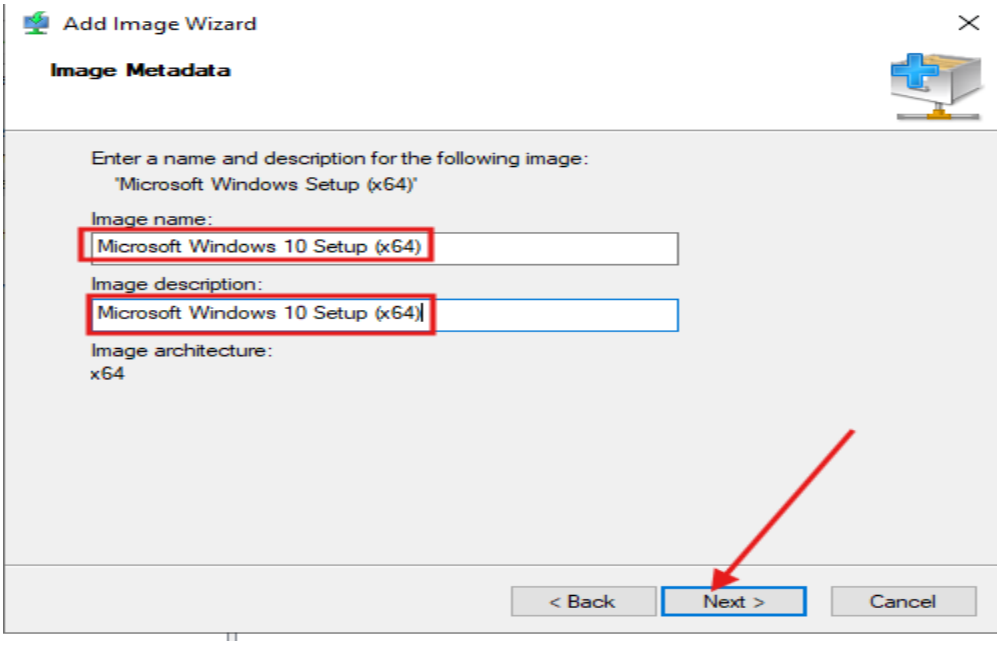
Go to the folder named Sources, select the boot.wim file, and click Next.

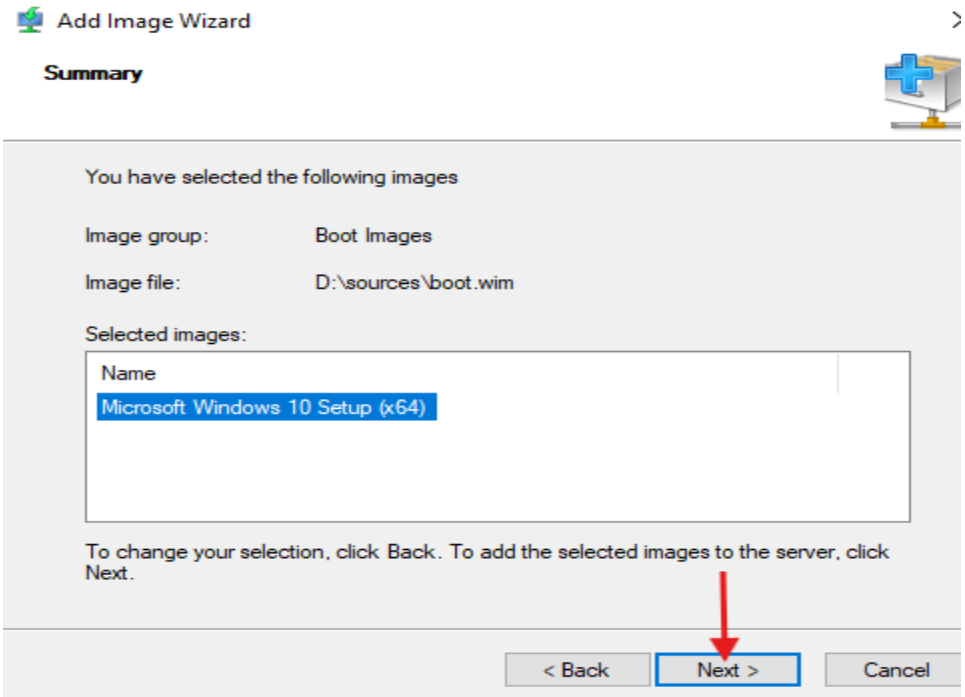


Here, you can see that the location has been specified.

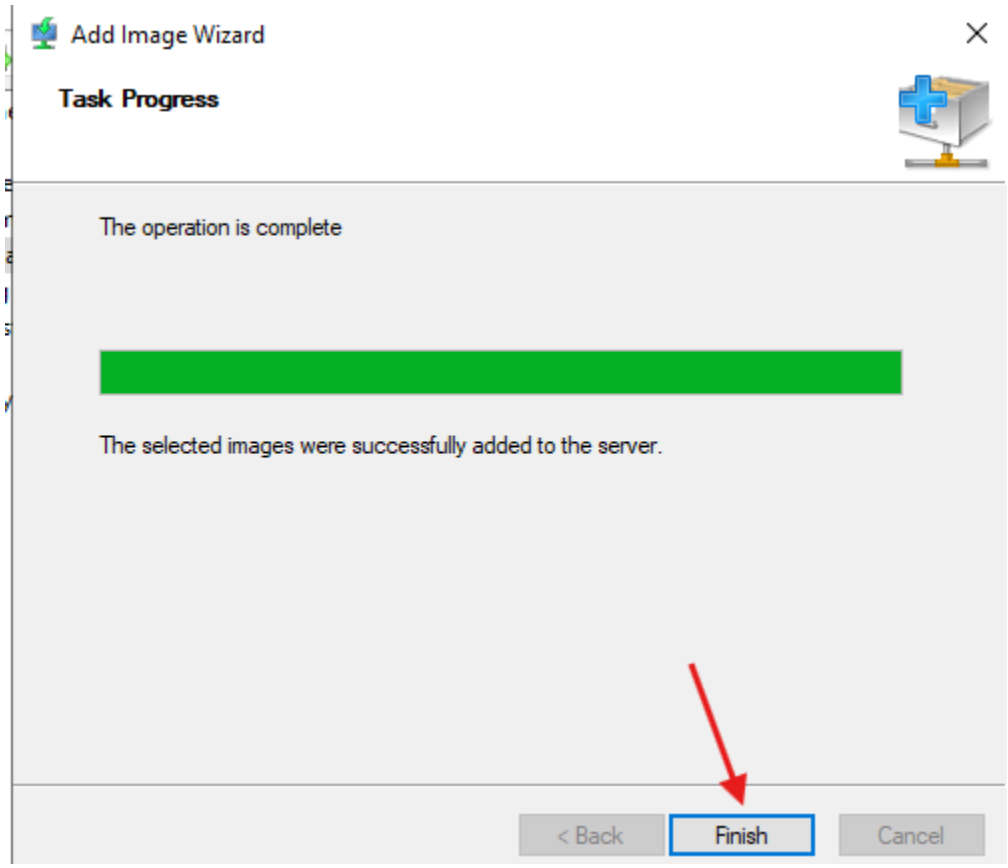


You can customize the image name and the description.

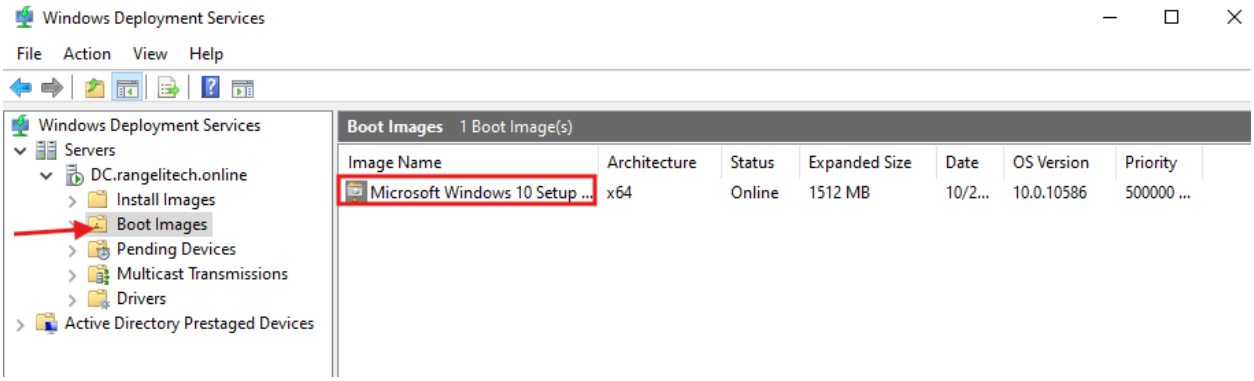




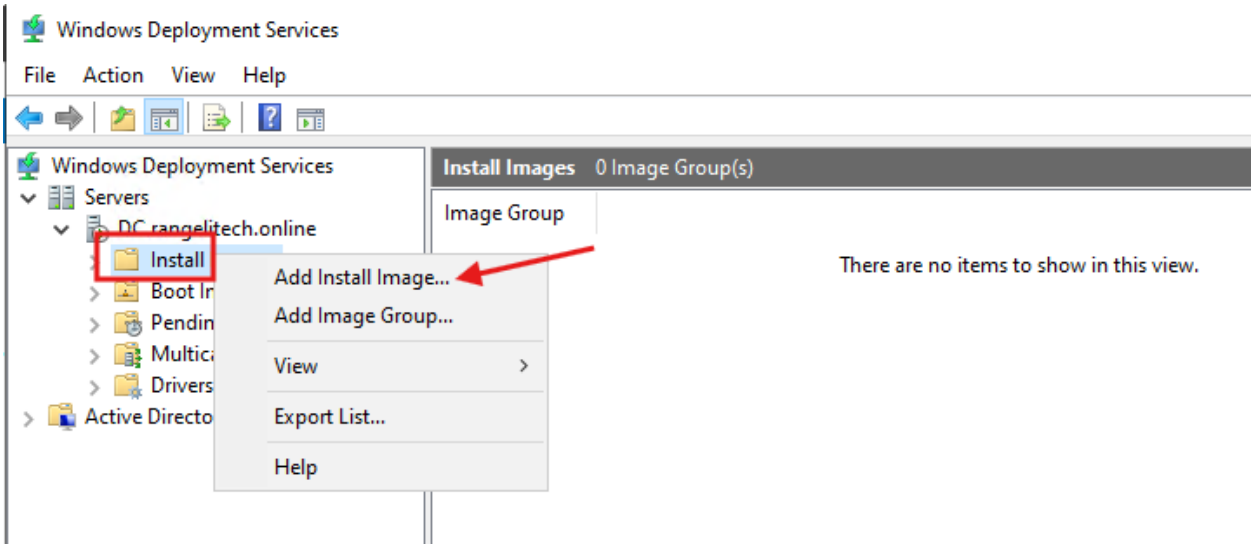
Now the operation is complete; the selected image has been successfully added to the WDS server. Let's click the Finish button to close this console.



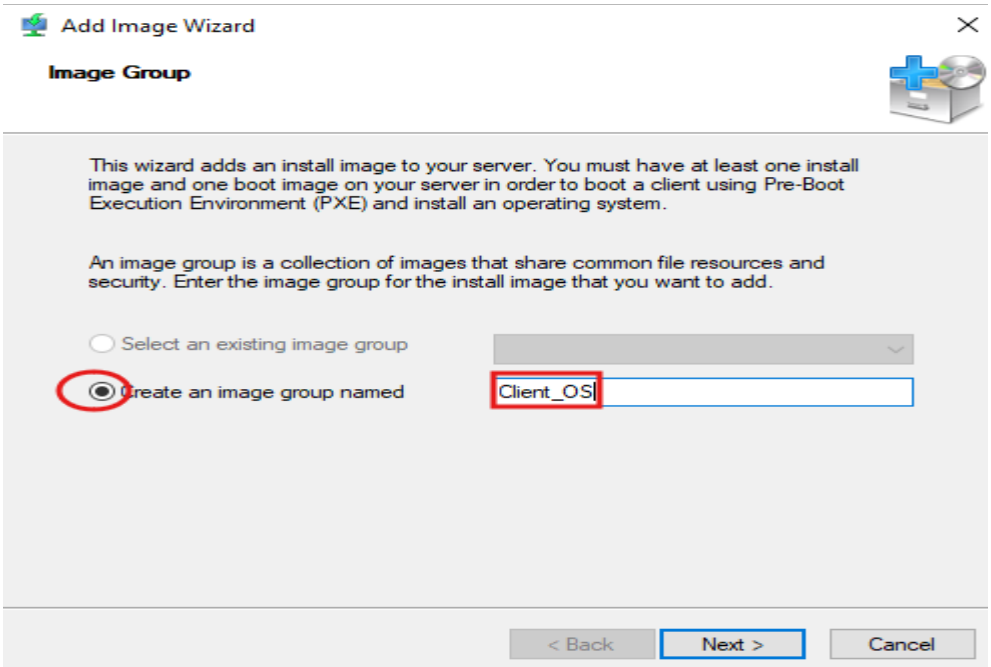
Now go to the WDS Management Console to verify that the boot image has been successfully added.



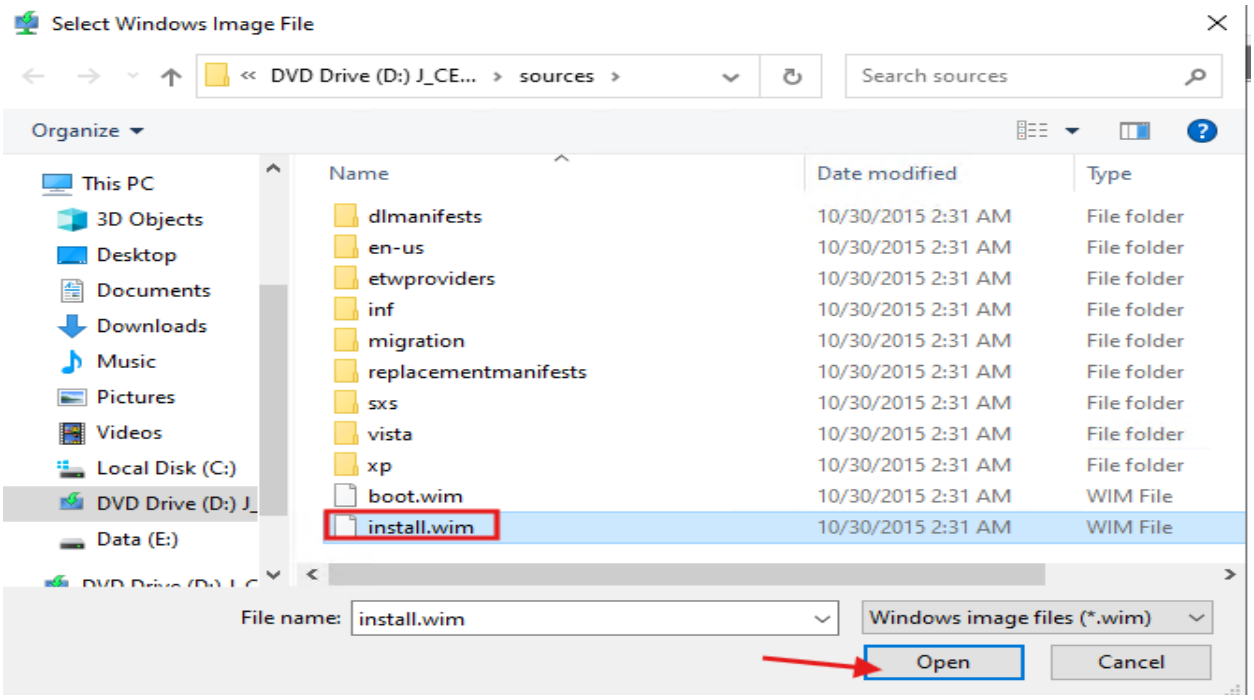
Now we are going to add an install image. Right-click on Install Images and select Add Install Image.



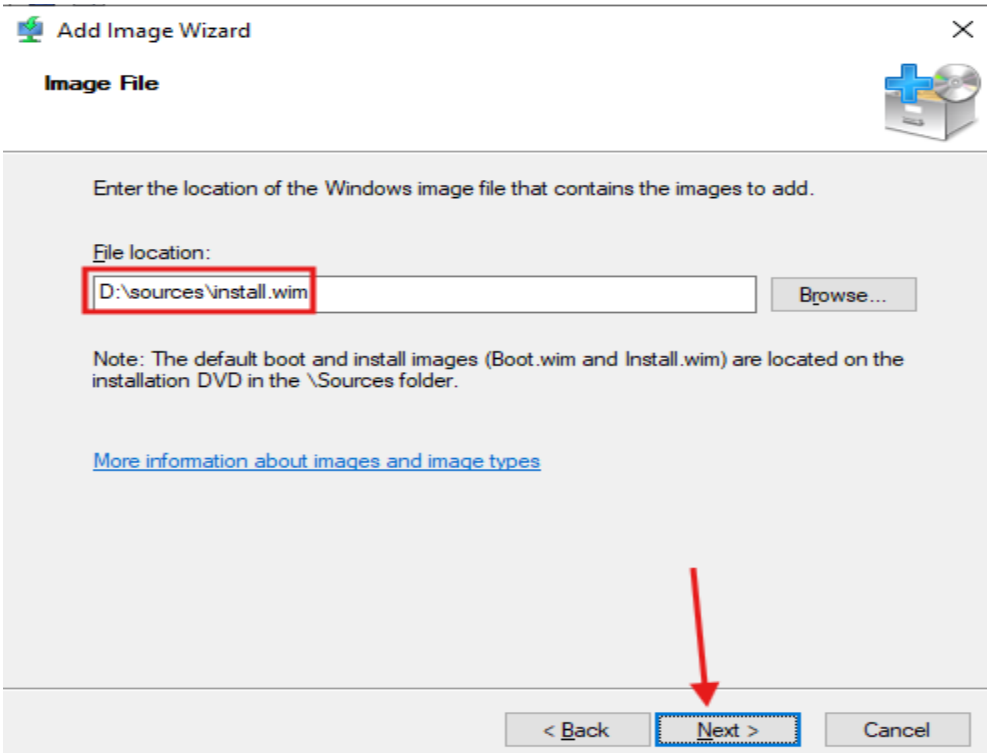
Here, let's give a name to the image group and hit Next.



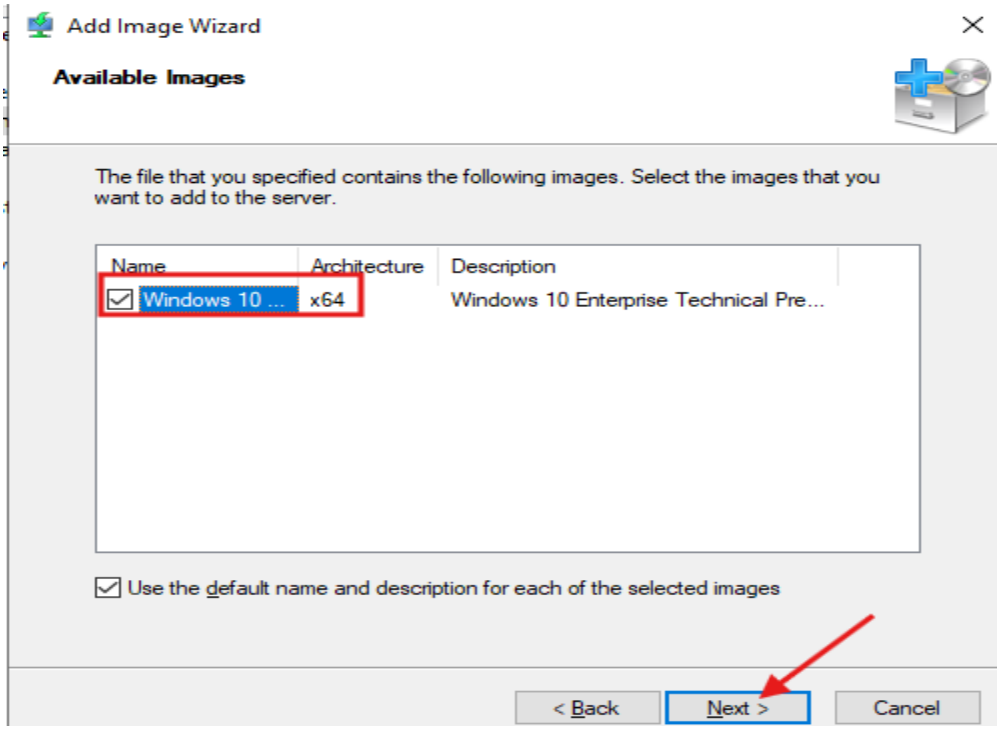
It will again ask you to specify the location, so click on Browse, specify the path, and select the install.wim file.

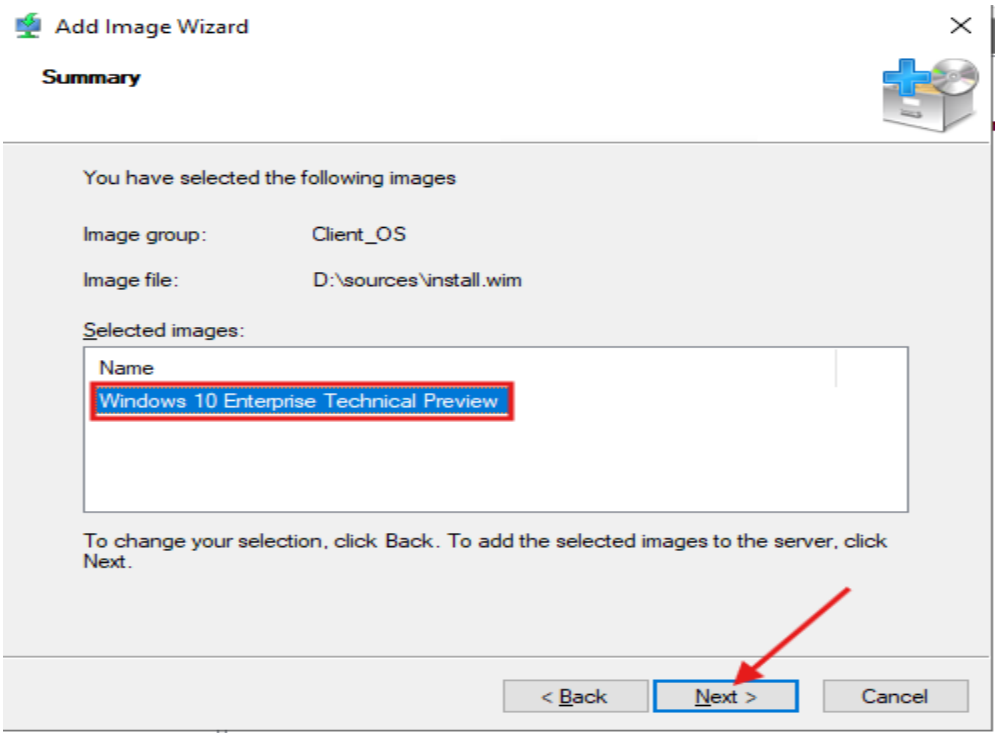


The location has been taken and hit Next.

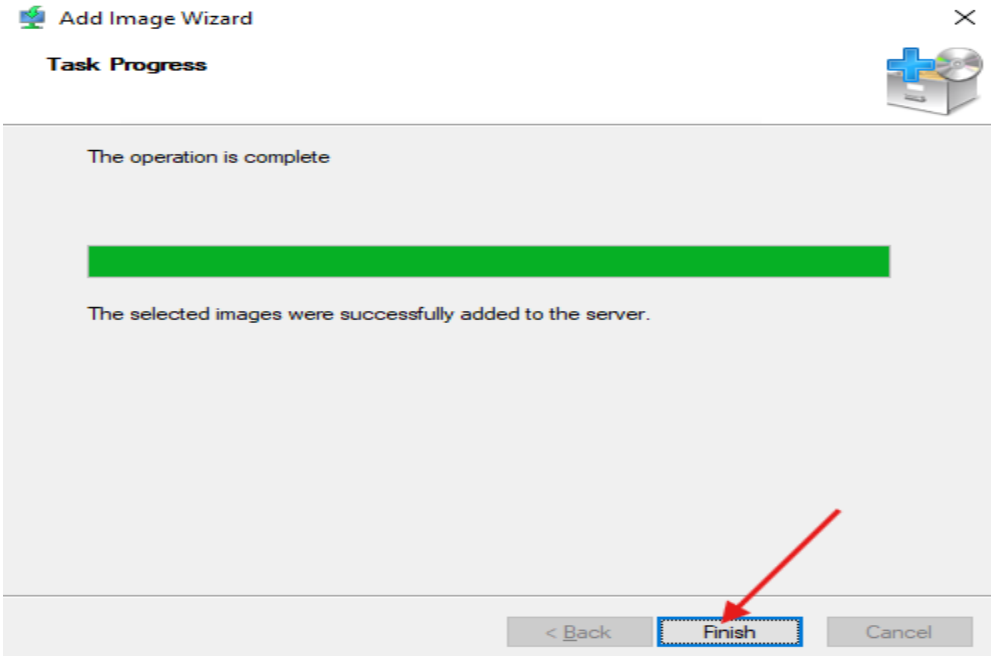


Here, you can see the Windows 10 ISO file and click Next.

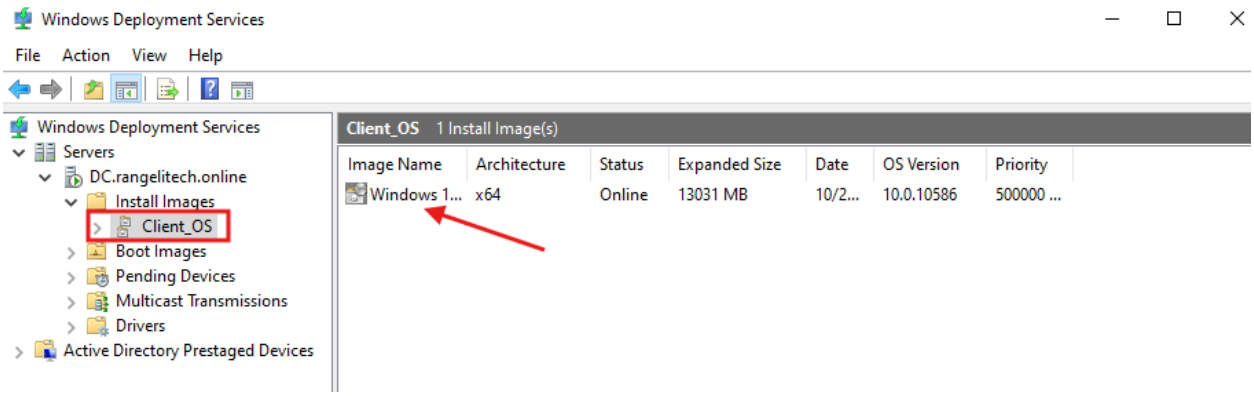




Here, it will take some time because the WDS server is copying the install.wim image from the installation media to the local server, after installation click Finish button.

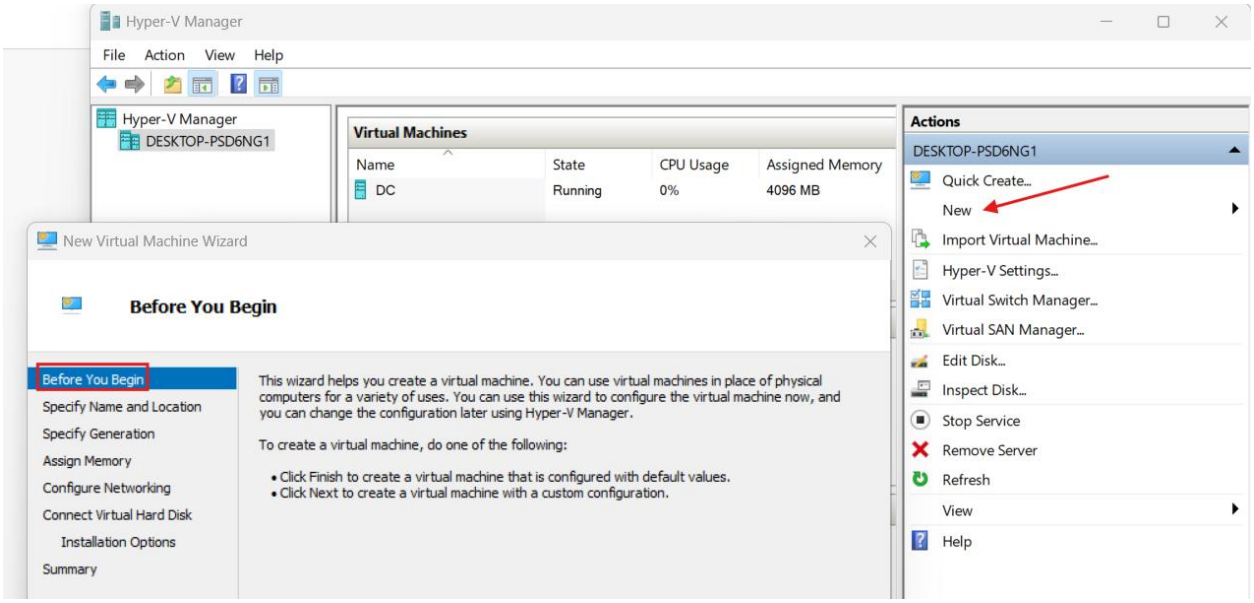


Now we can verify that the image group is Client_OS, and under that, there is the Windows 10 image.

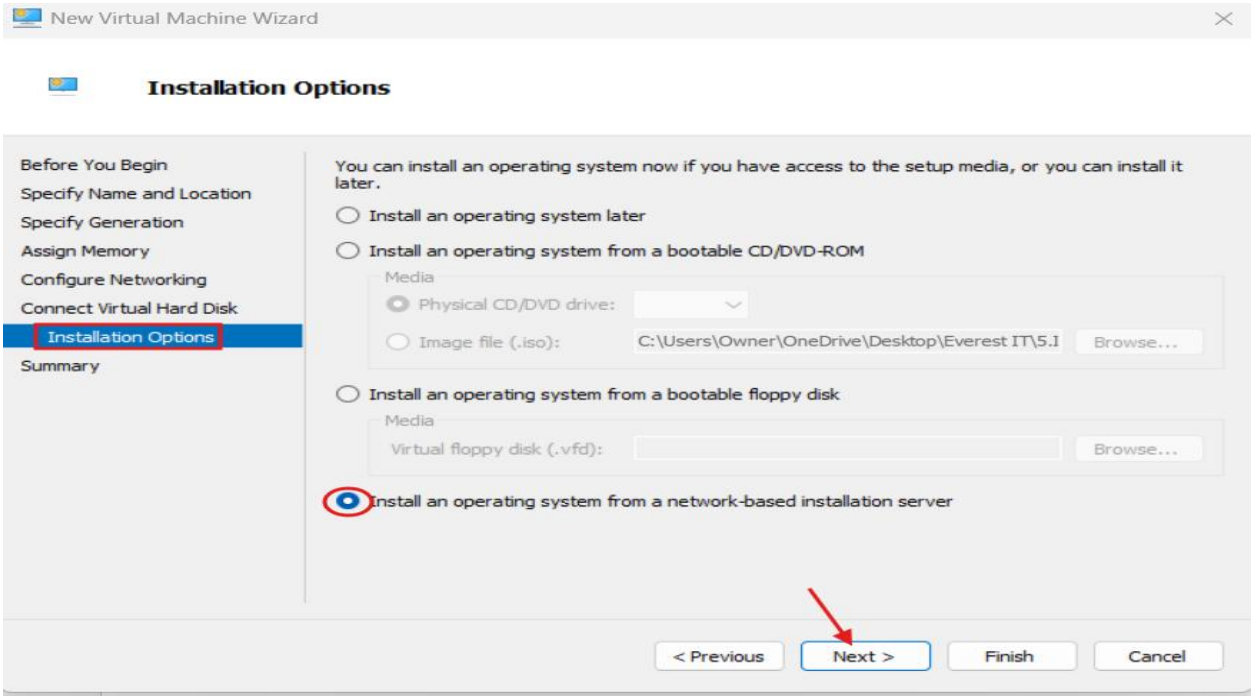


Finally, we are done with the basic configurations of WDS. Now you can use the WDS server to deploy the Windows 10 operating system to our network devices using WDS services.

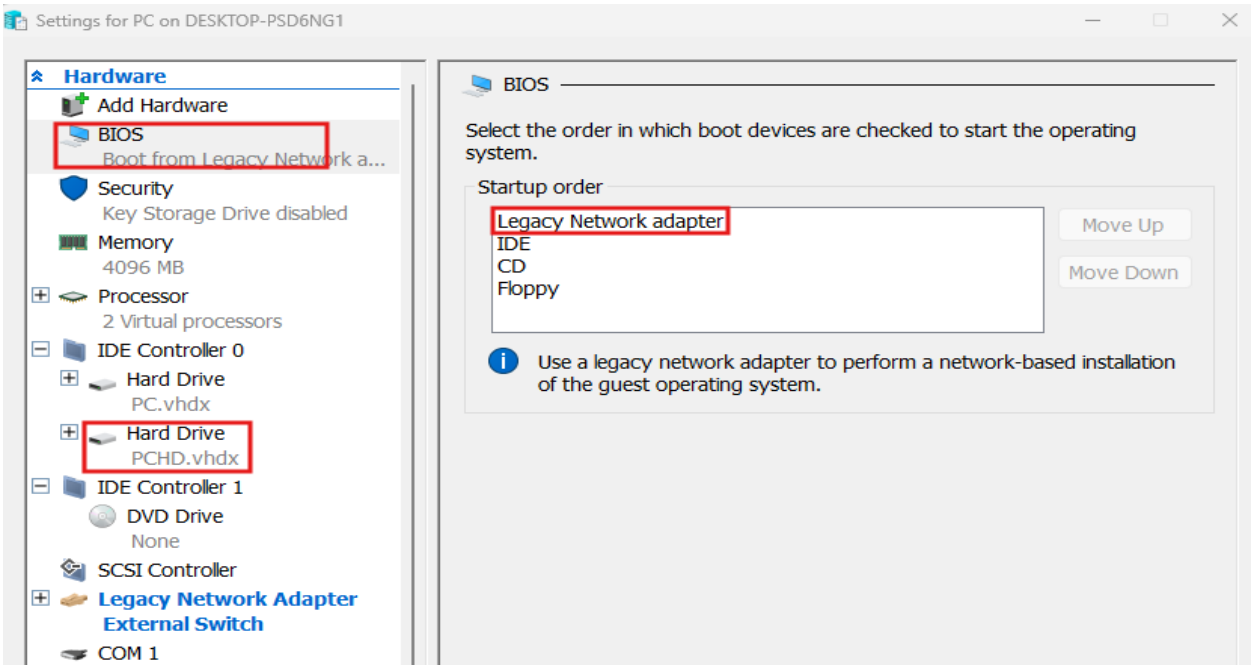
Now go to Hyper-V Manager and create a virtual machine.



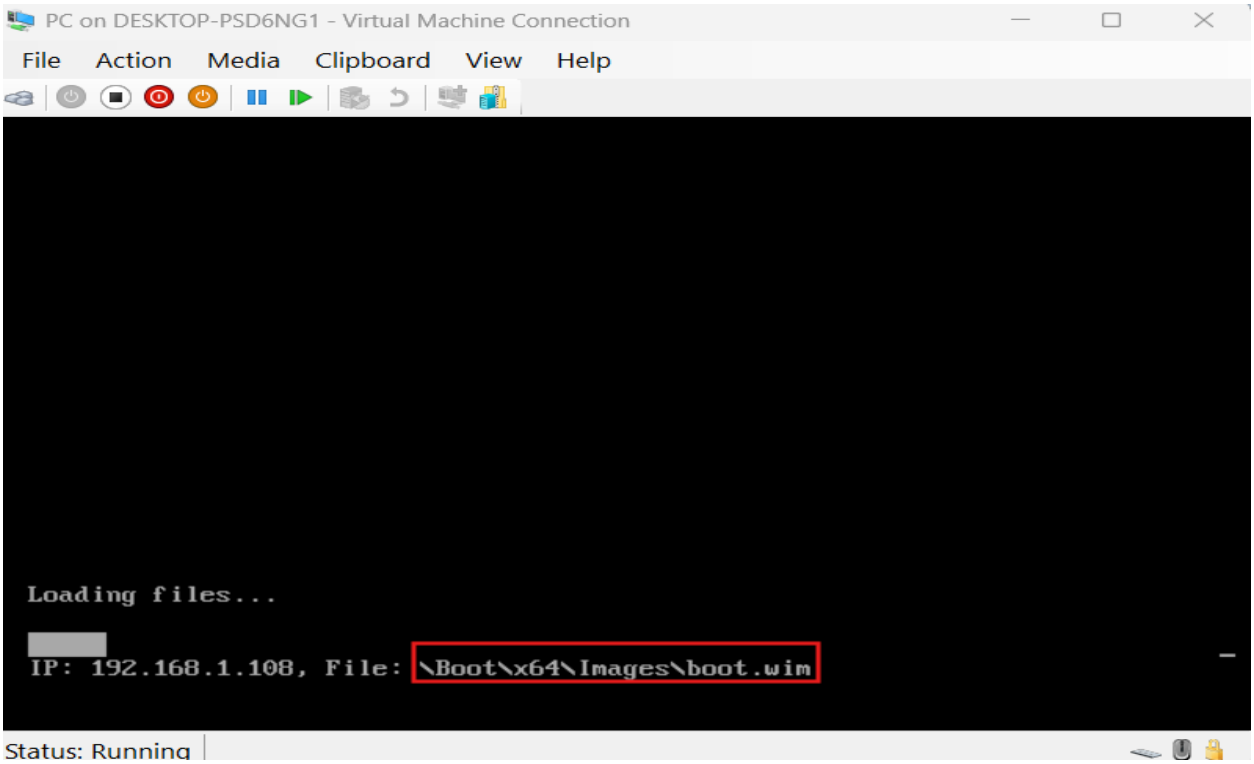
Here, in Installation Options, select Install an operating system from a network-based installation server because we want to boot this VM using the network.



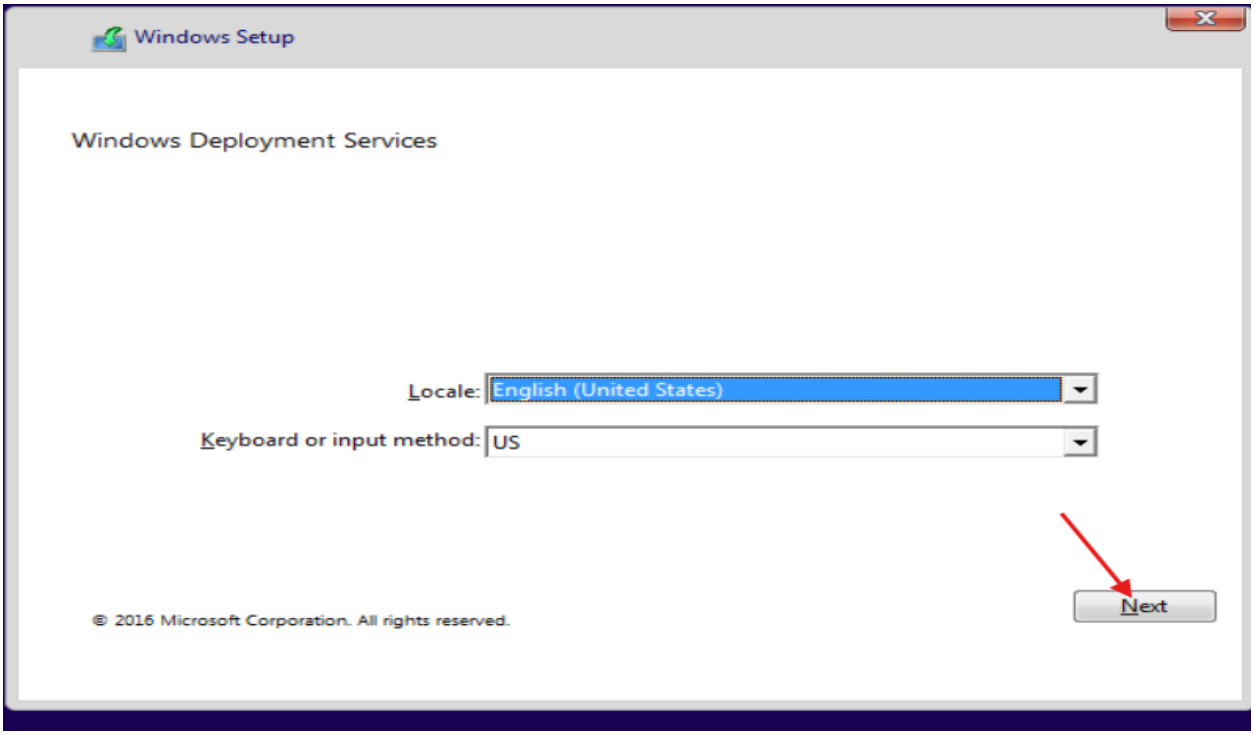
You can do some housekeeping on the VM settings. Go to BIOS and arrange the startup order as shown in the screenshot. Additionally, add a hard drive that you will need to save the captured image files.



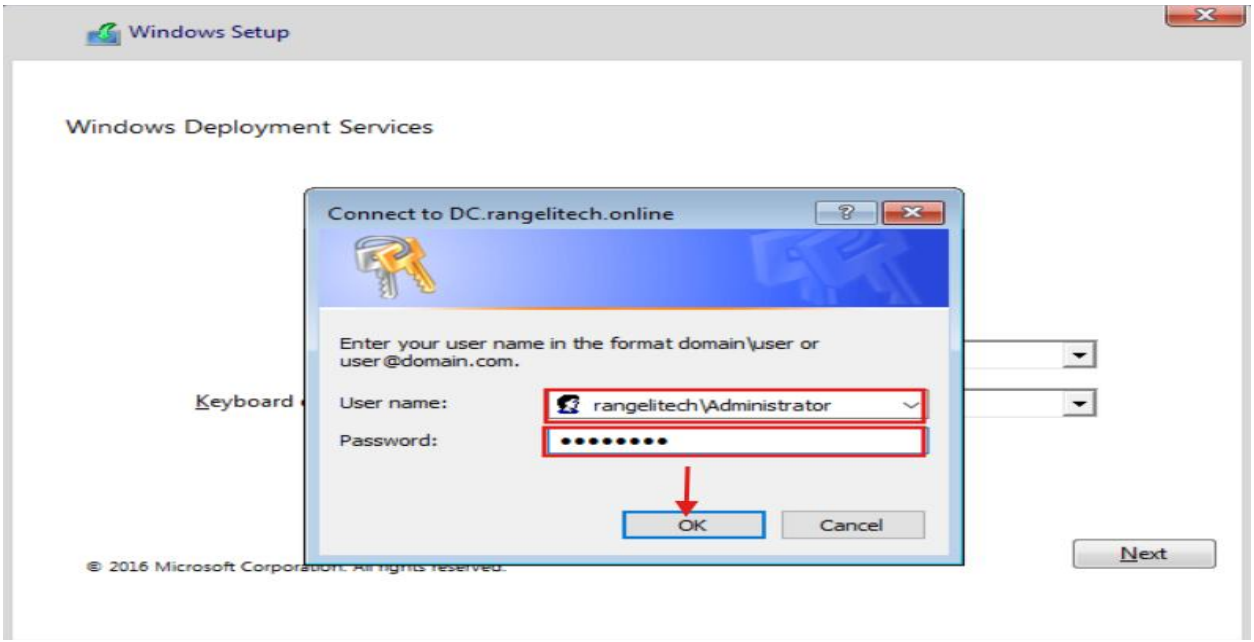
Let's start with the VM, Once the VM starts, you need to press the F12 key for network boot. The client machine loads the boot.wim from the WDS server.



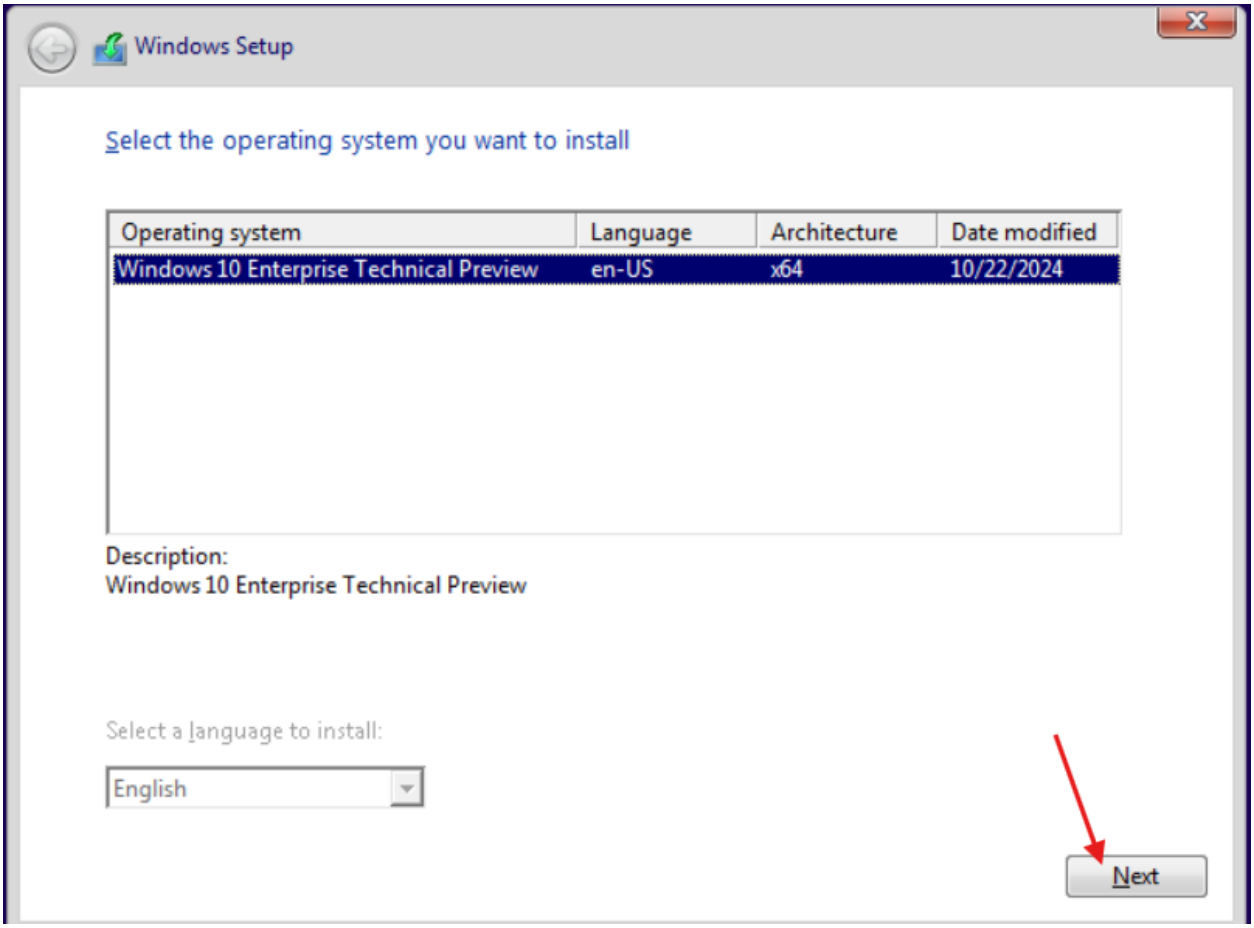
In this wizard, you can choose the applicable option and click Next.



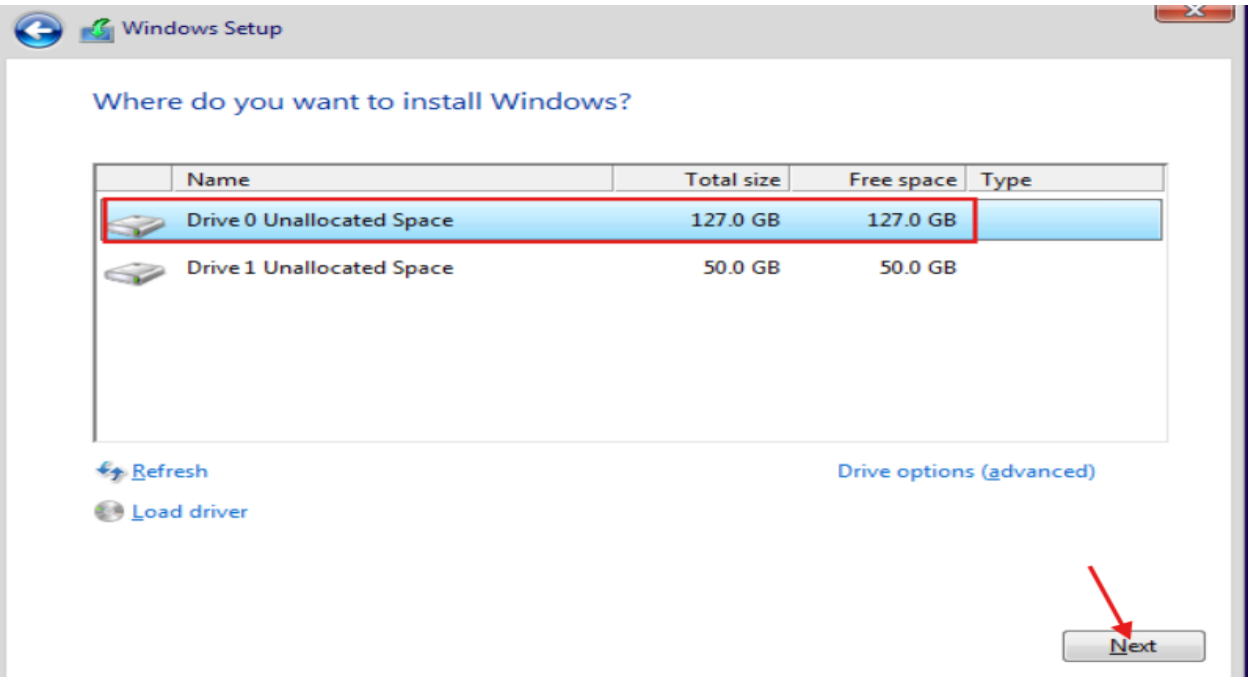
Here, you will be asked to enter the server credentials. Enter them and click OK.



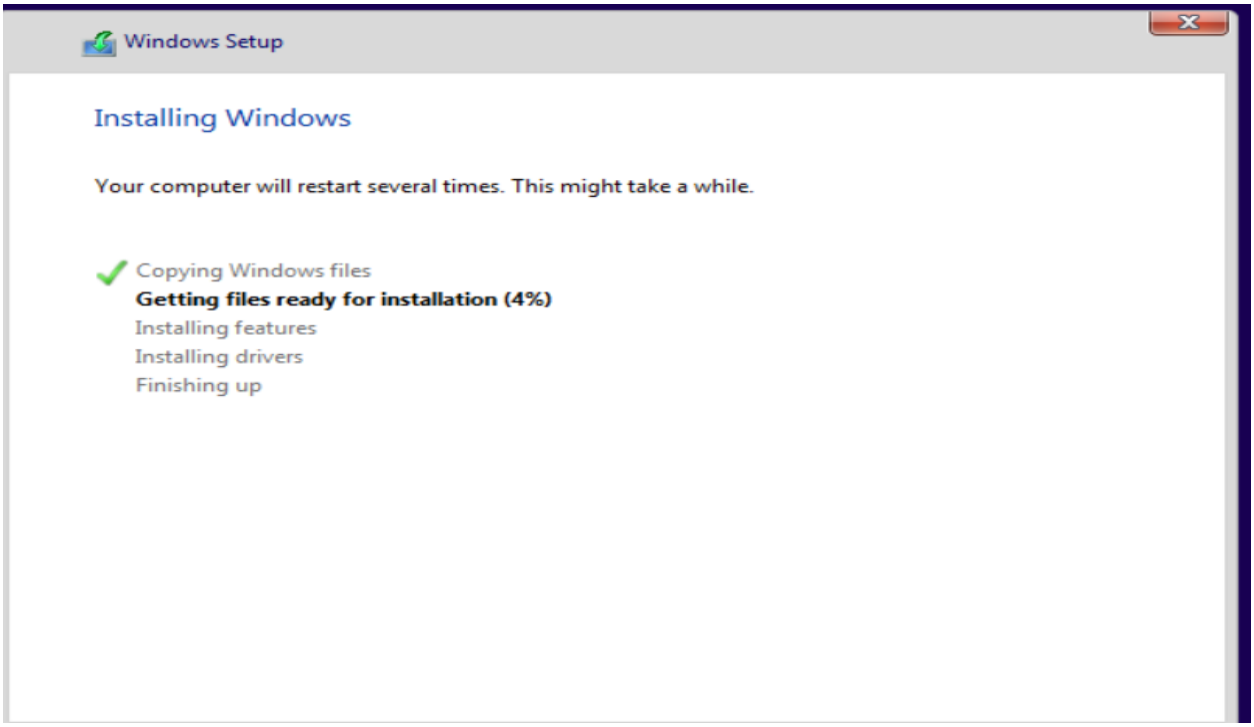
Here, you will see the installation image for the Windows 10 operating system. Click Next.



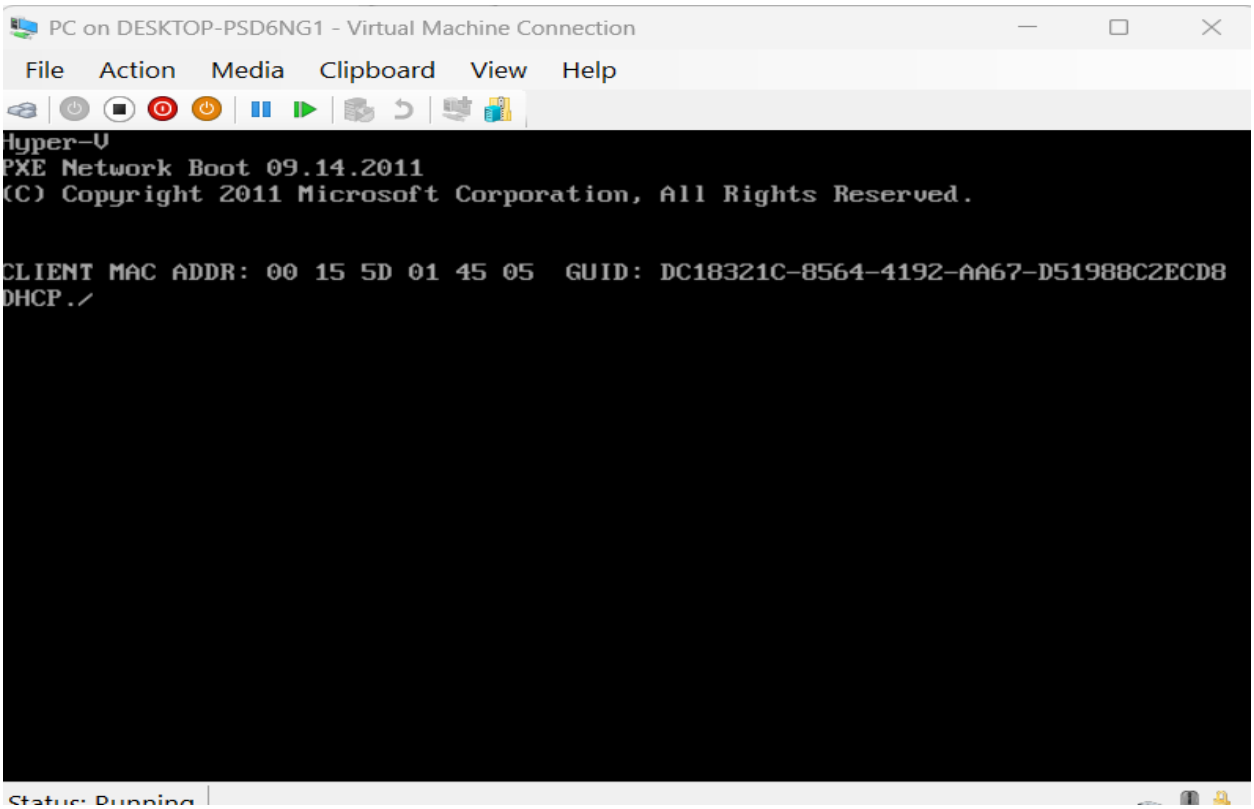
Here, choose the appropriate drive option and click Next.



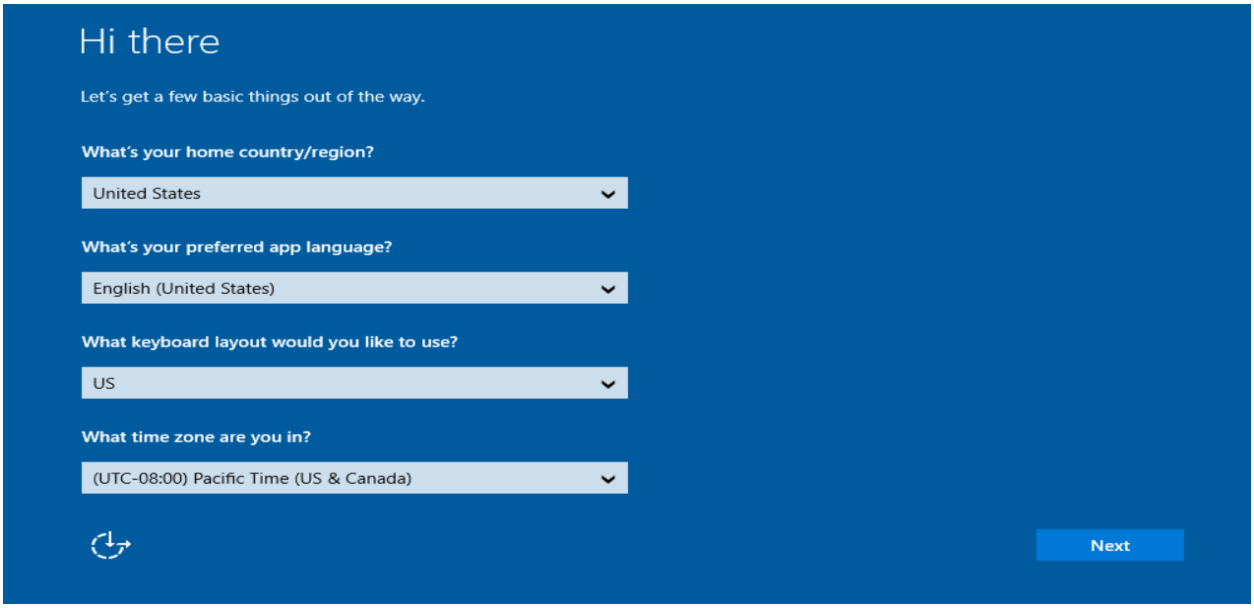
Now, the Windows installation process will start.



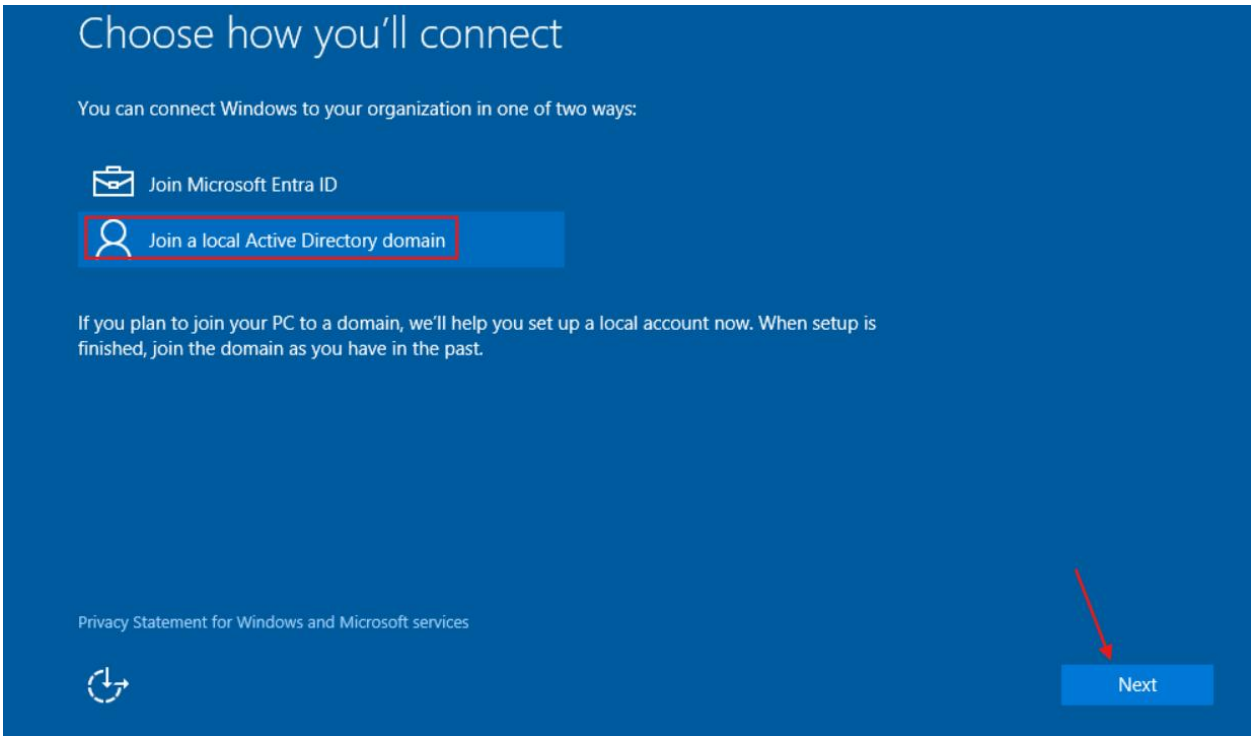
Once the installation is completed, you will see this screen, but don't press the F12 key; otherwise, your Windows 10 client will start up again using boot.wim from the WDS server.



After the VM installation, you will see this window. Check the required fields and click Next.

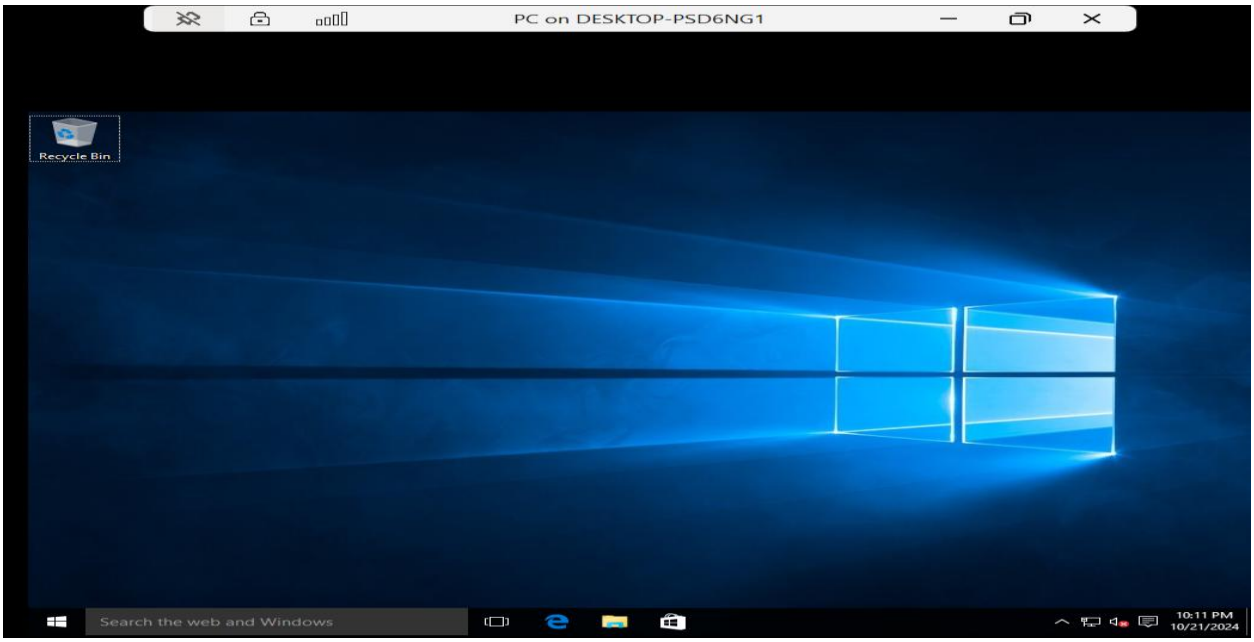


Select the options join a local Active Directory domain and click Next.



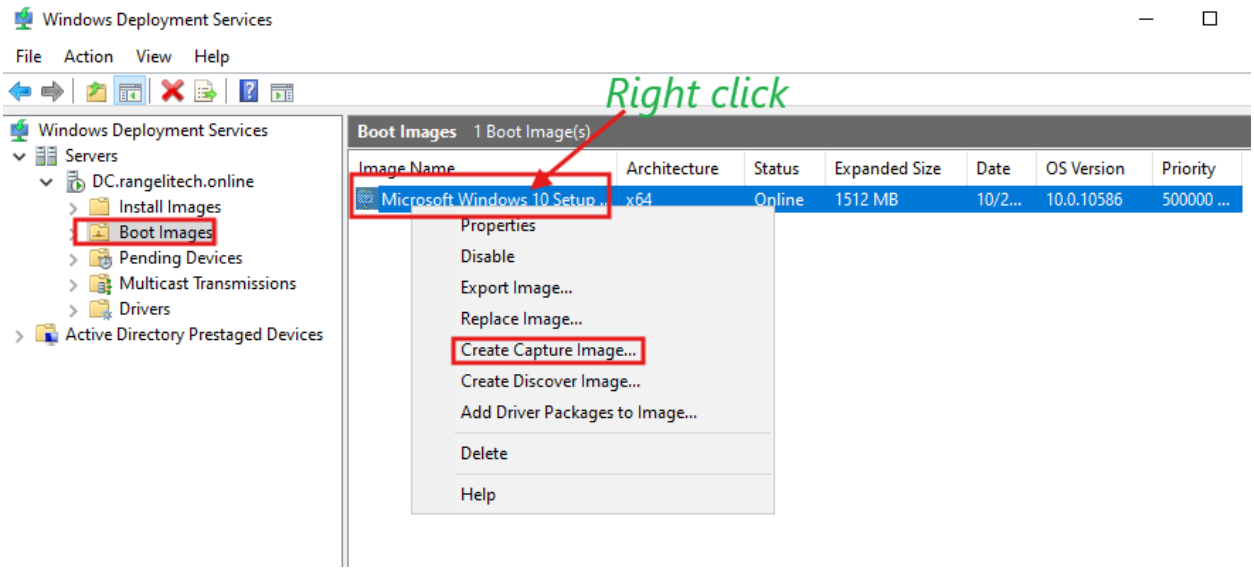
Create an account for this VM and click Next.

Now, we have a newly installed desktop Windows 10 client computer.

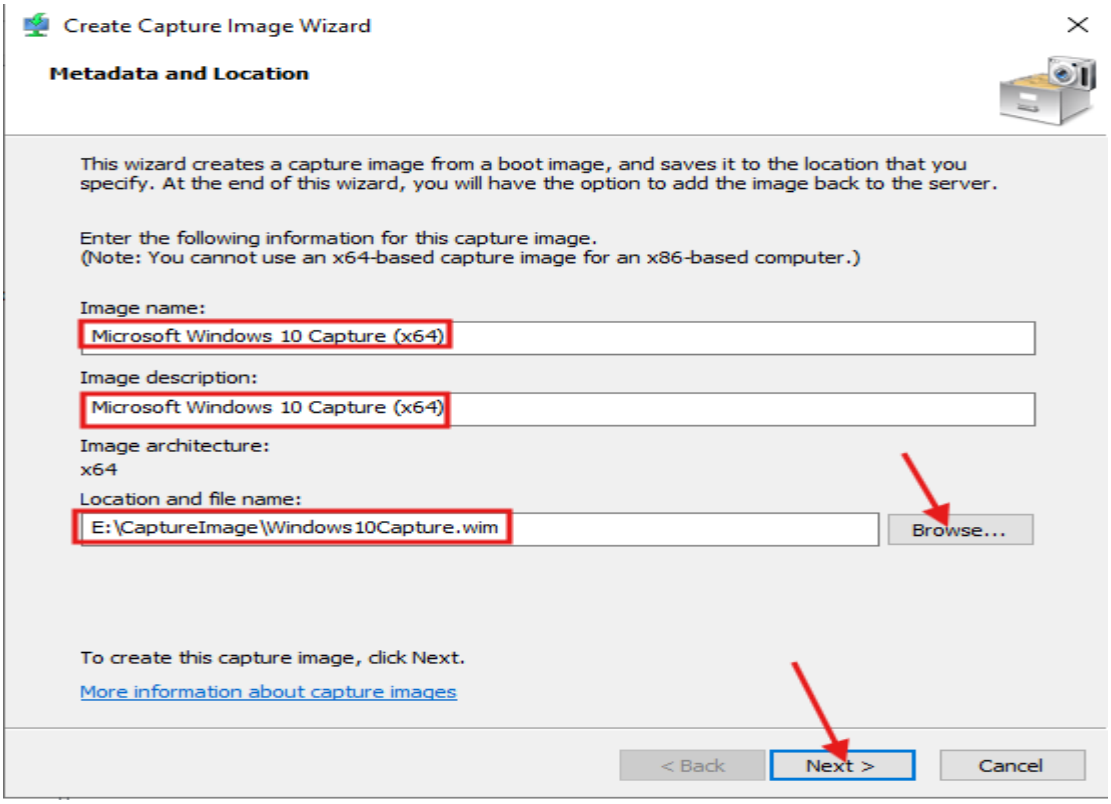


Steps create a captured image from an existing boot image in Windows Deployment Services (WDS).

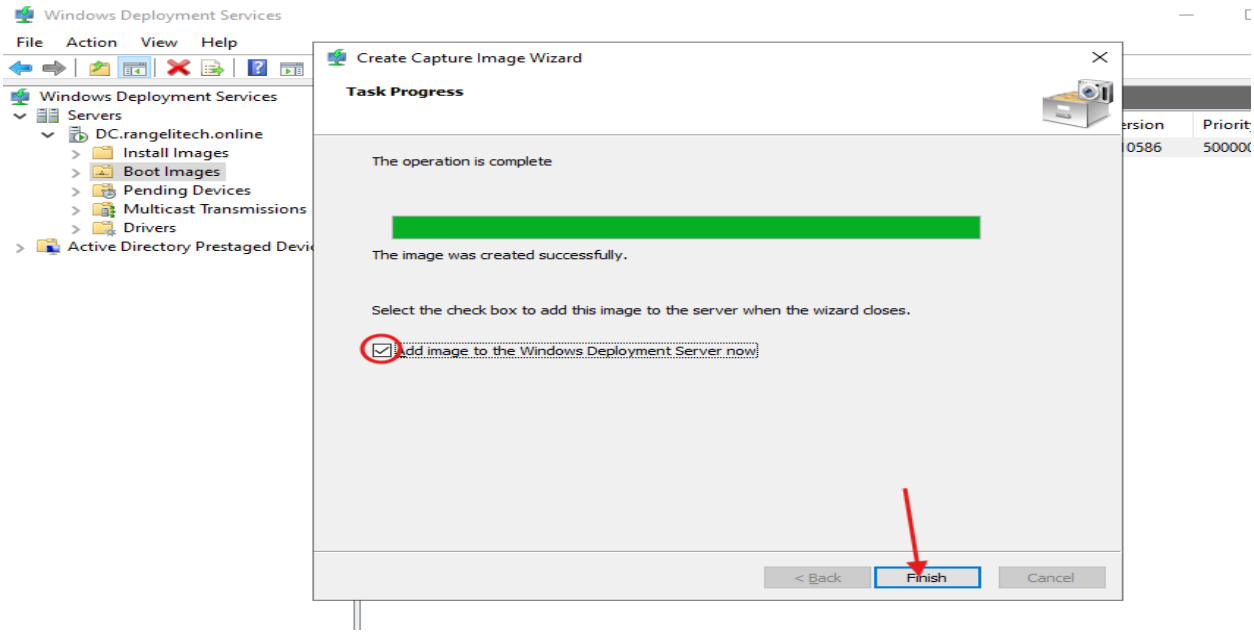
Let's go to the WDS Management Console. Click on Boot Images, right-click on the boot image, and select Create Computer Image.



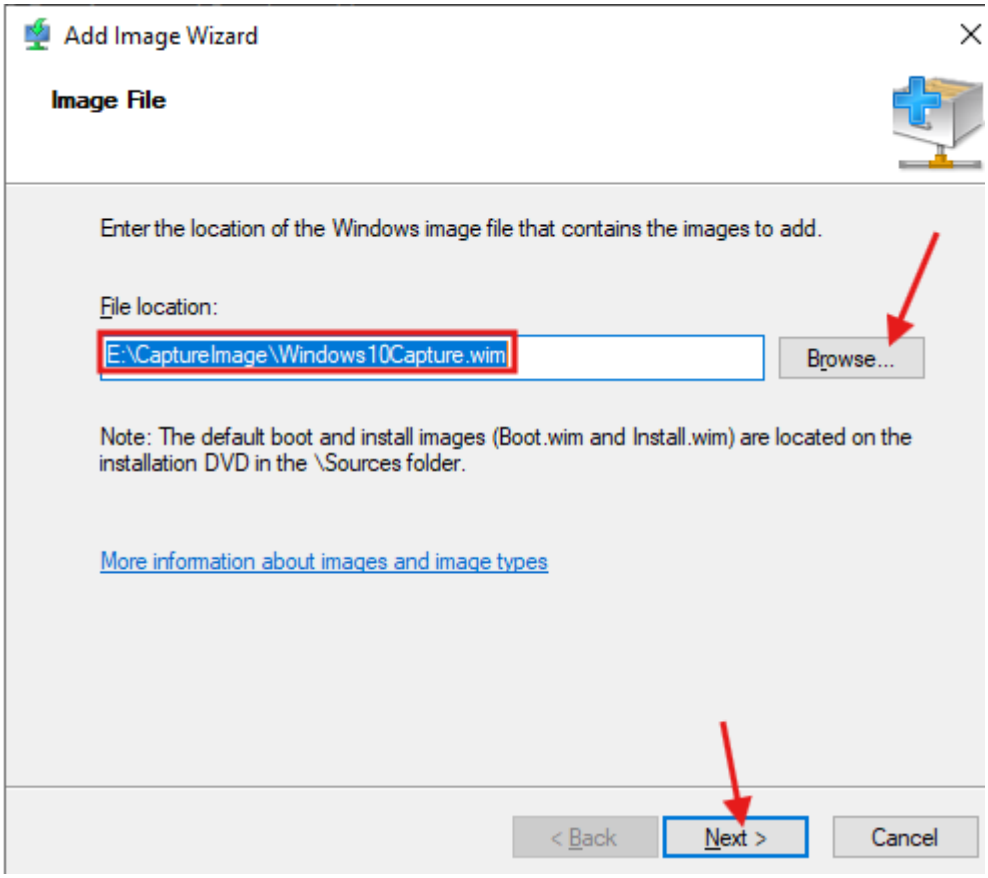
Here, you need to specify the name and description of the image, as well as the location.



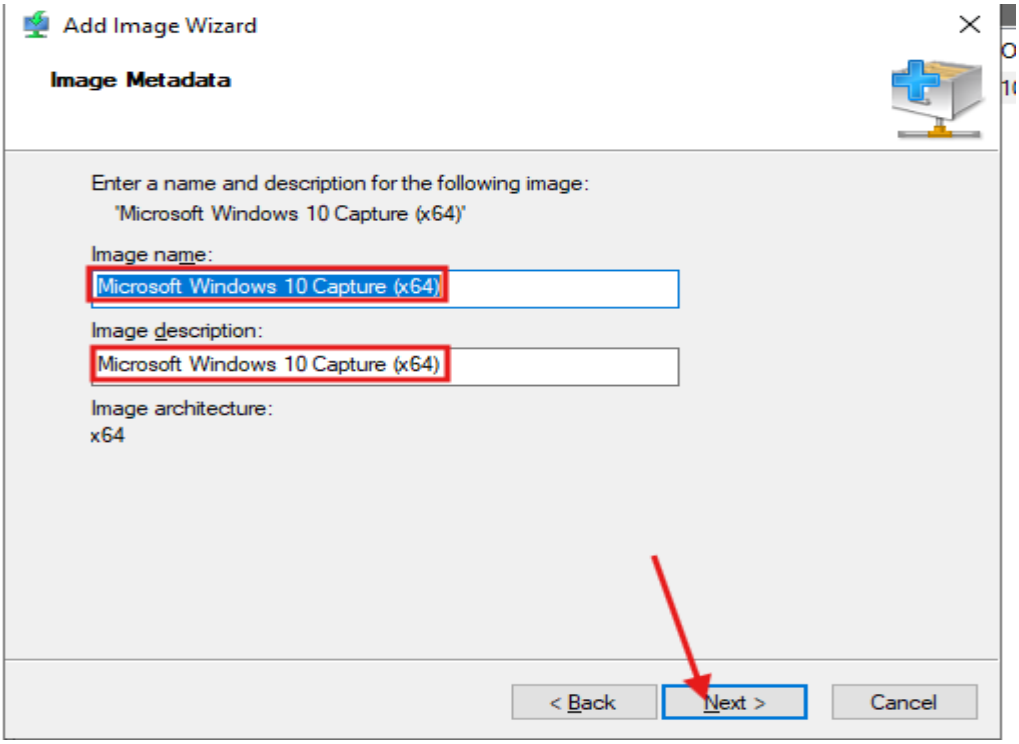
It will capture the image for you and save it to the location you specified. Now, you need to add this image to the WDS server and select the check box Add image and click Finish.



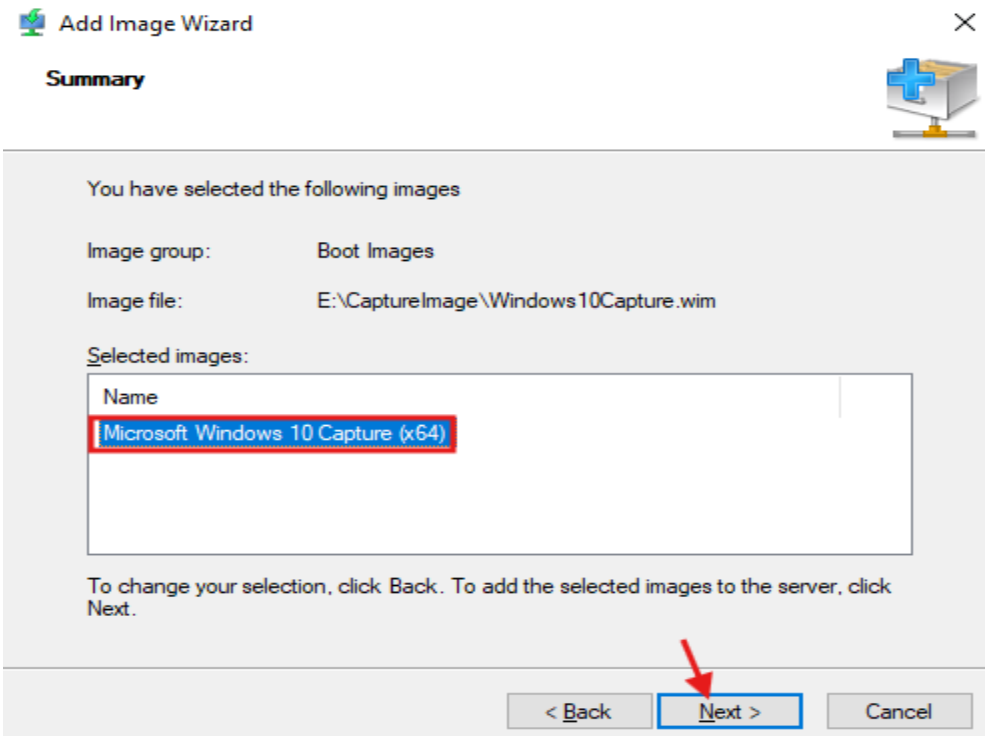
The captured image has been saved to the specified location.



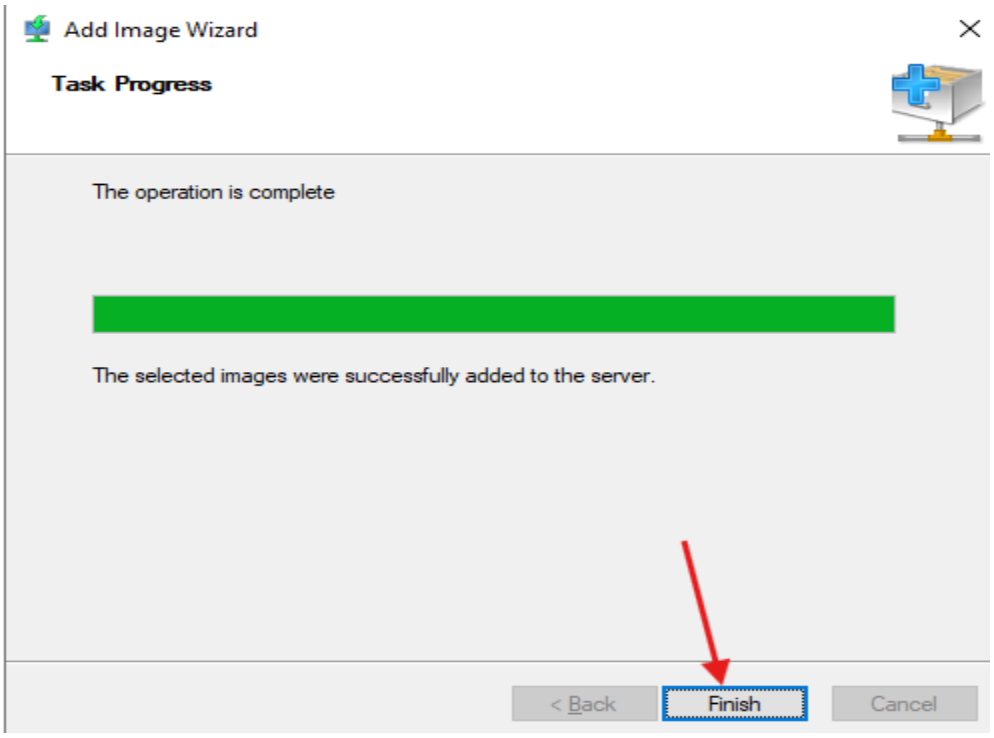
This will be the name and description of the captured image.

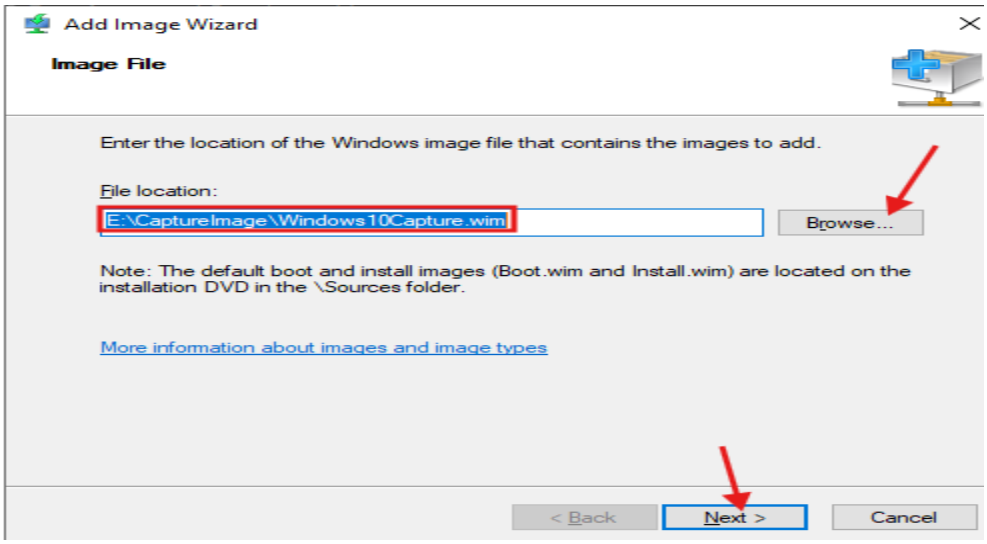


Click Next.

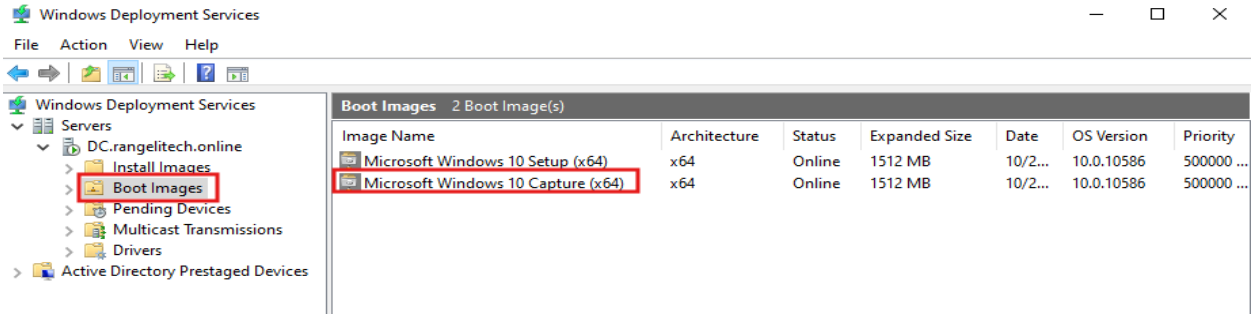


The capture has been saved to the WDS server.

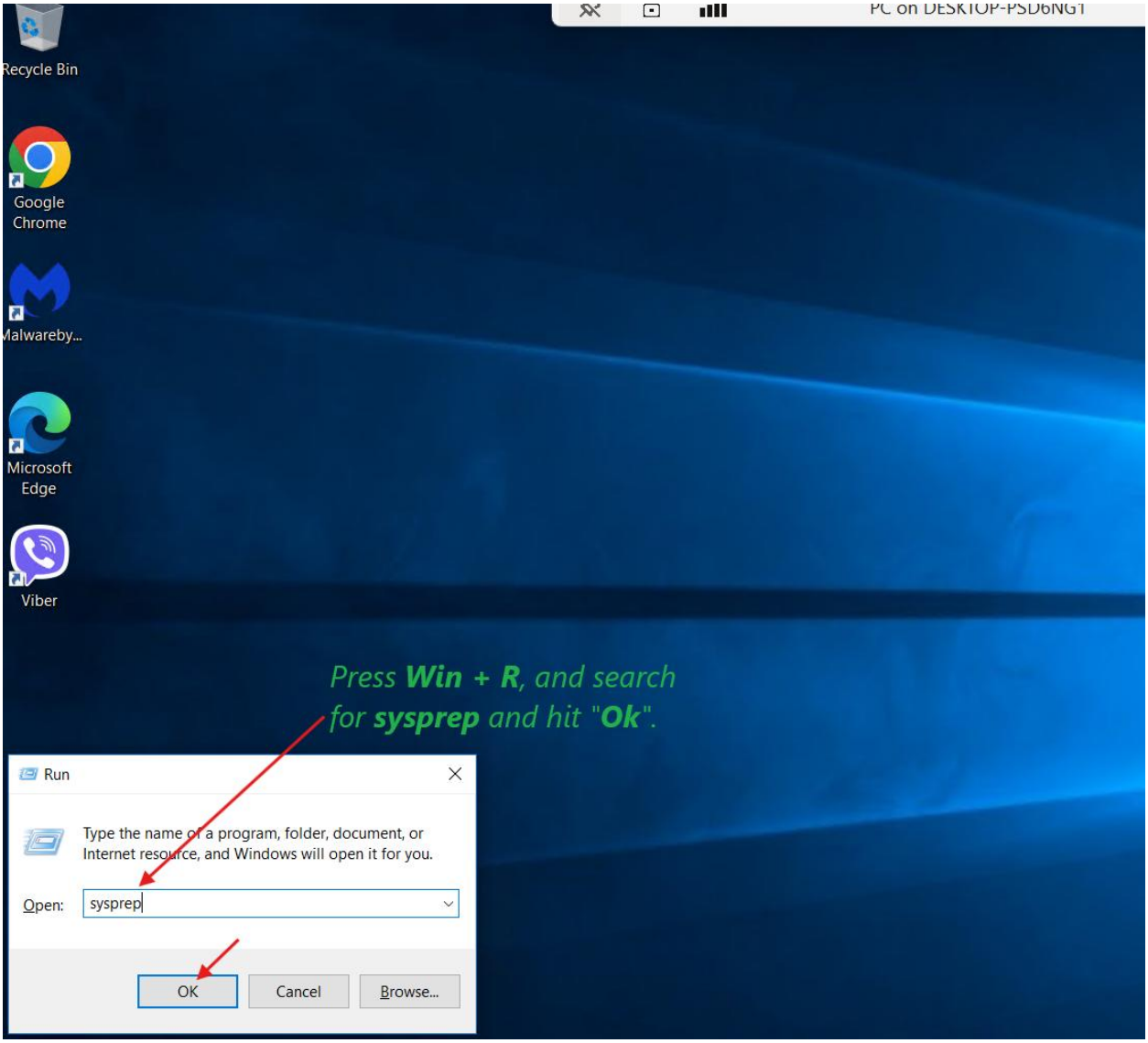




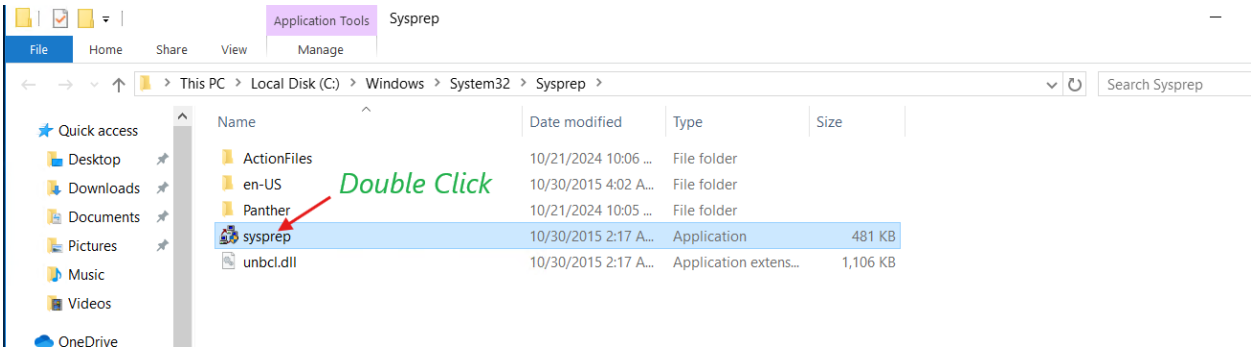
Here, you can verify the total images: the first is the regular Windows 10 boot image, and the second is the custom image you will use to capture the Windows 10 reference image. Now the server-side configuration is completed.



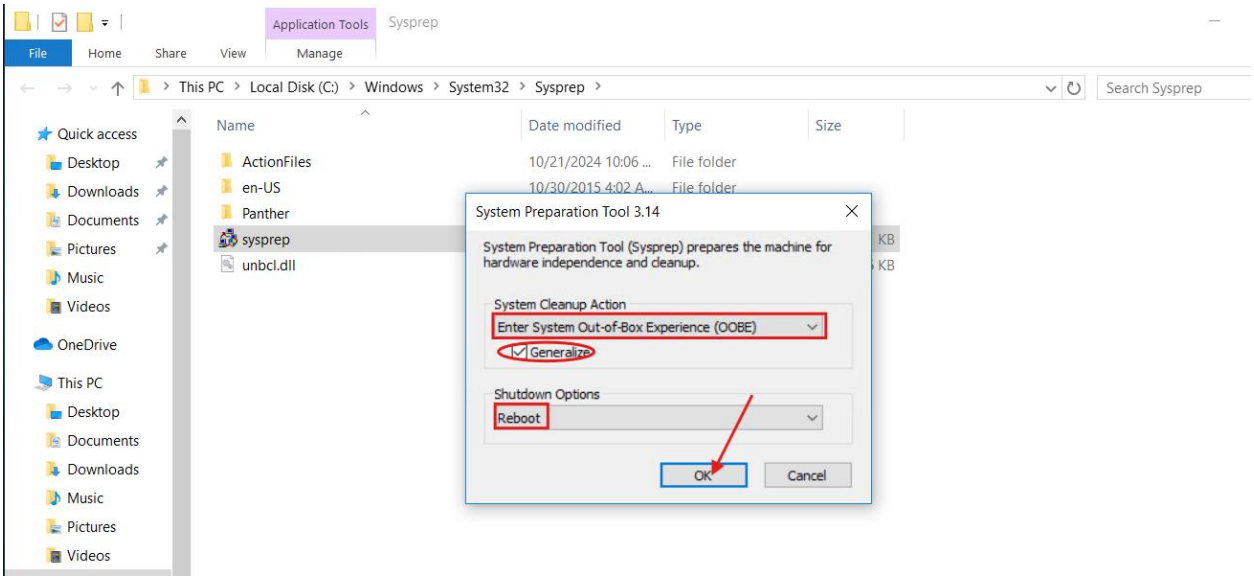
Now, let's move to the Windows 10 computer that we deployed earlier. On this computer, I have installed some applications. You can install additional applications according to your organization's requirements. Now, let's prepare this Windows 10 computer. Press Win + R and search for "sysprep."

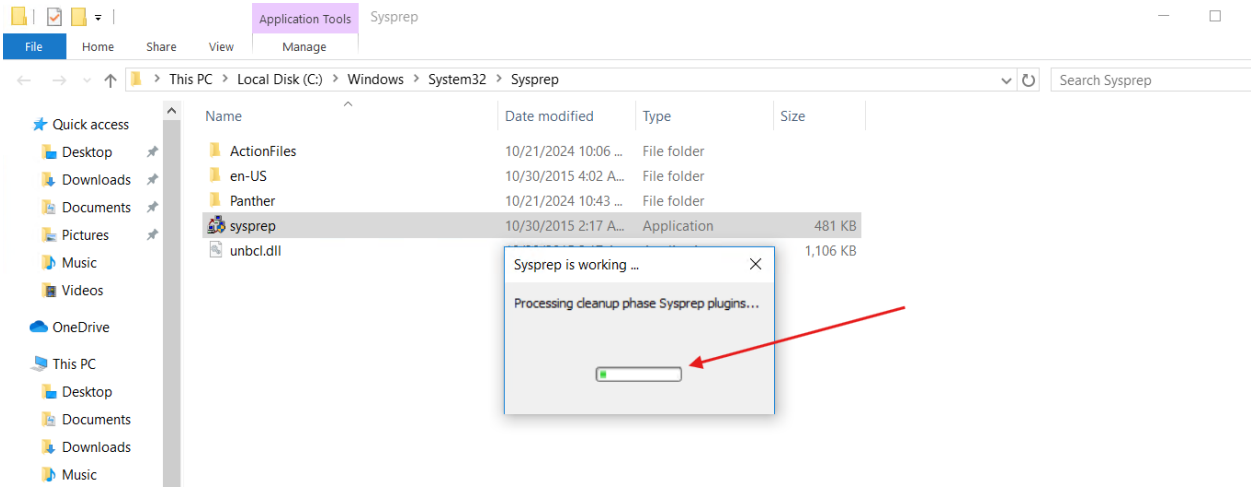


Here, double click on 'sysprep' application.

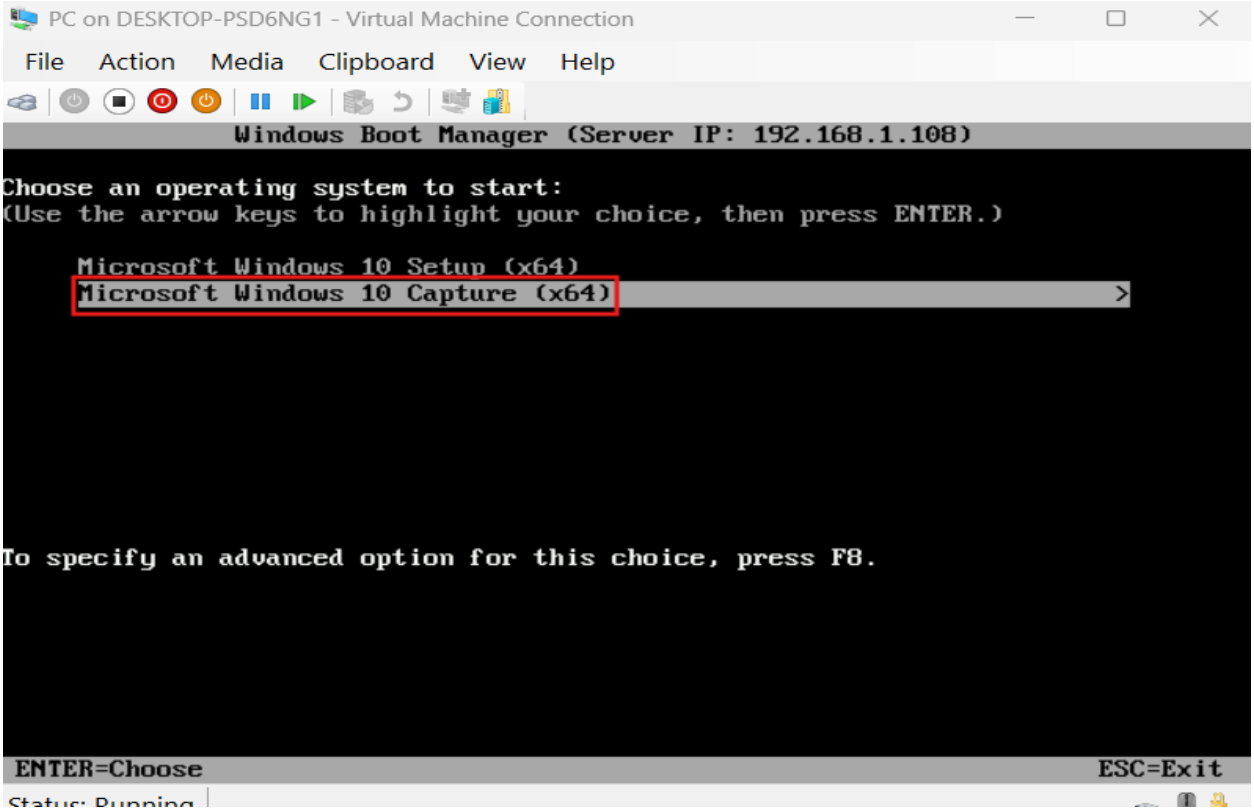


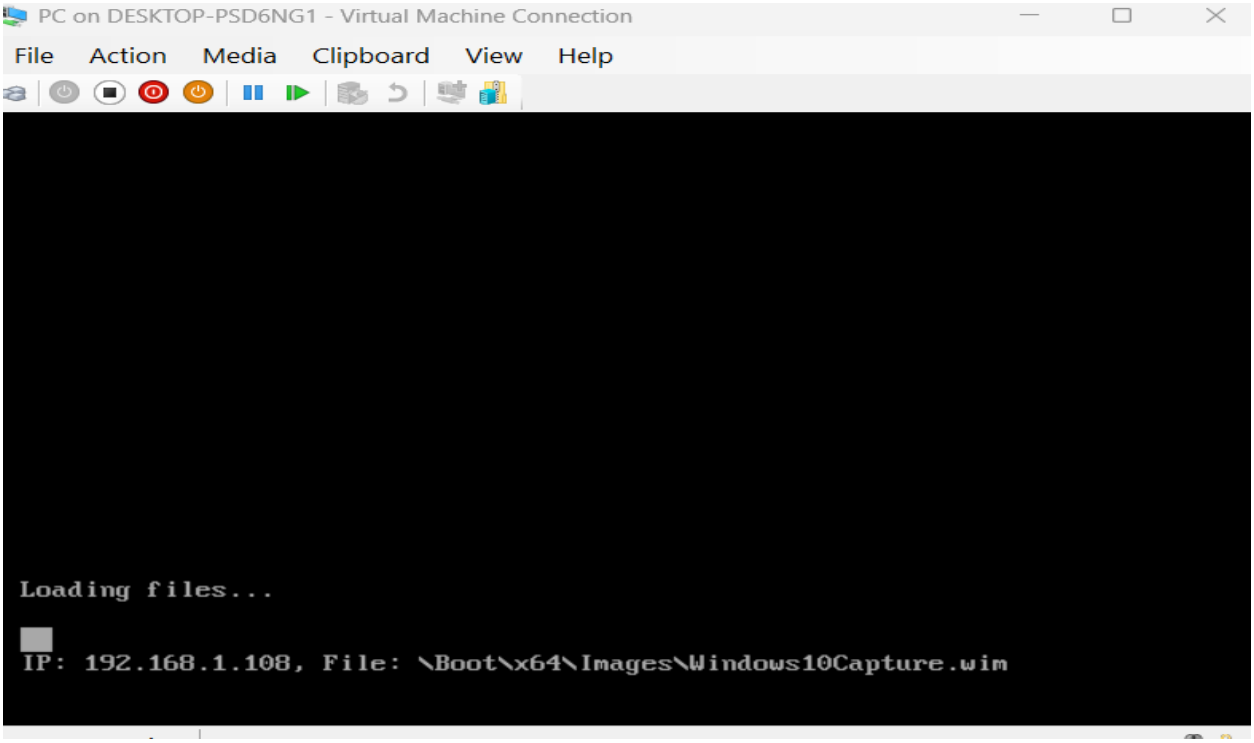
It will open the System Preparation Tool. This application is used to define your computer's name and generate a unique SID for the newly built computer.



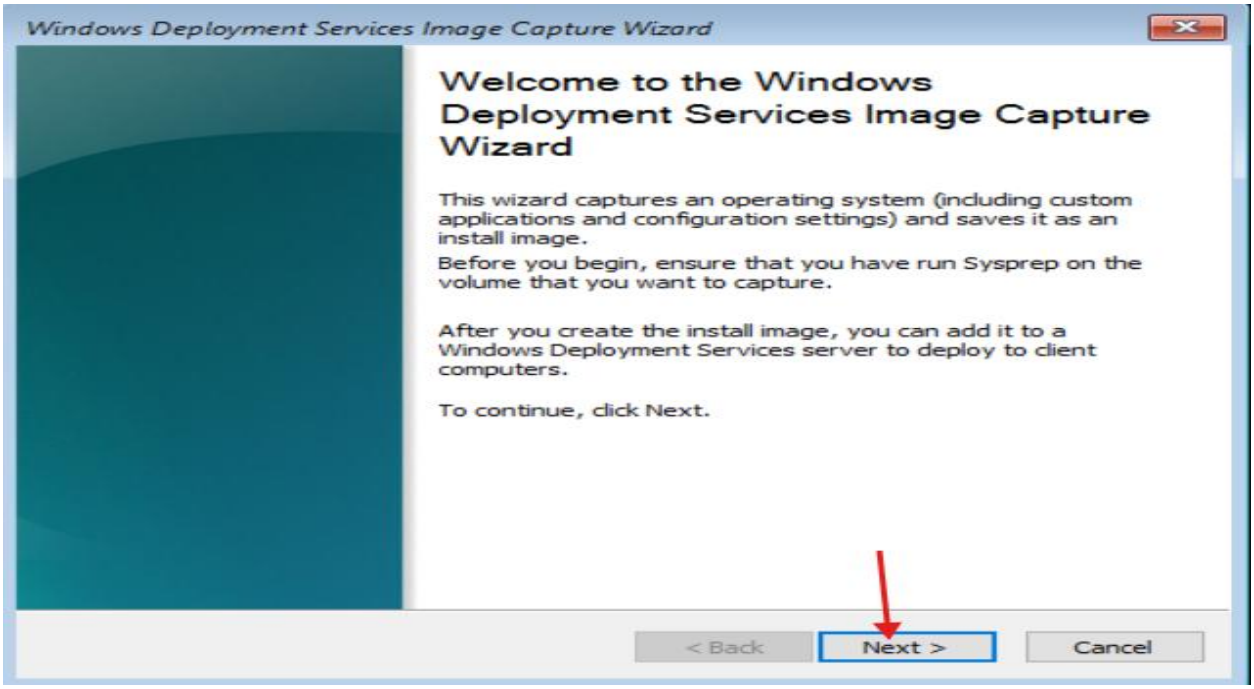


Now, this computer is configured to boot from the network. During the next boot, simply press F12 and select the 'Microsoft Windows 10 Capture (x64)' option, then press the Enter key.

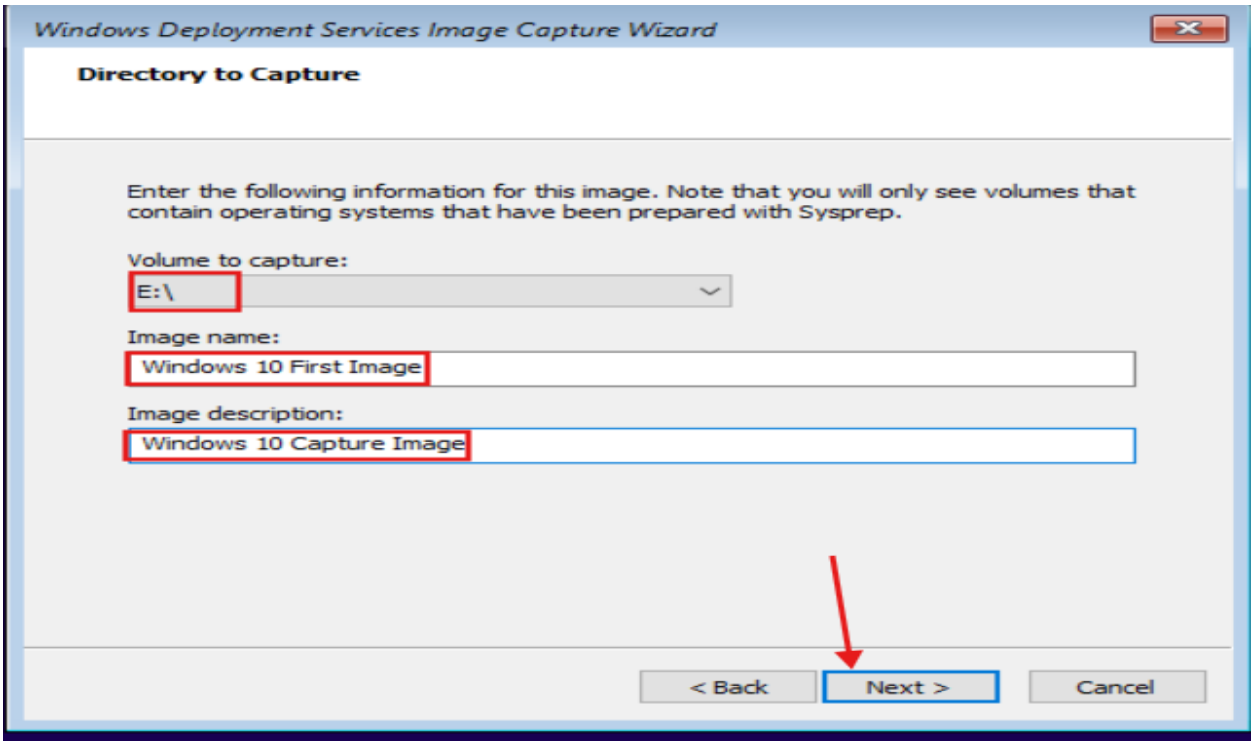




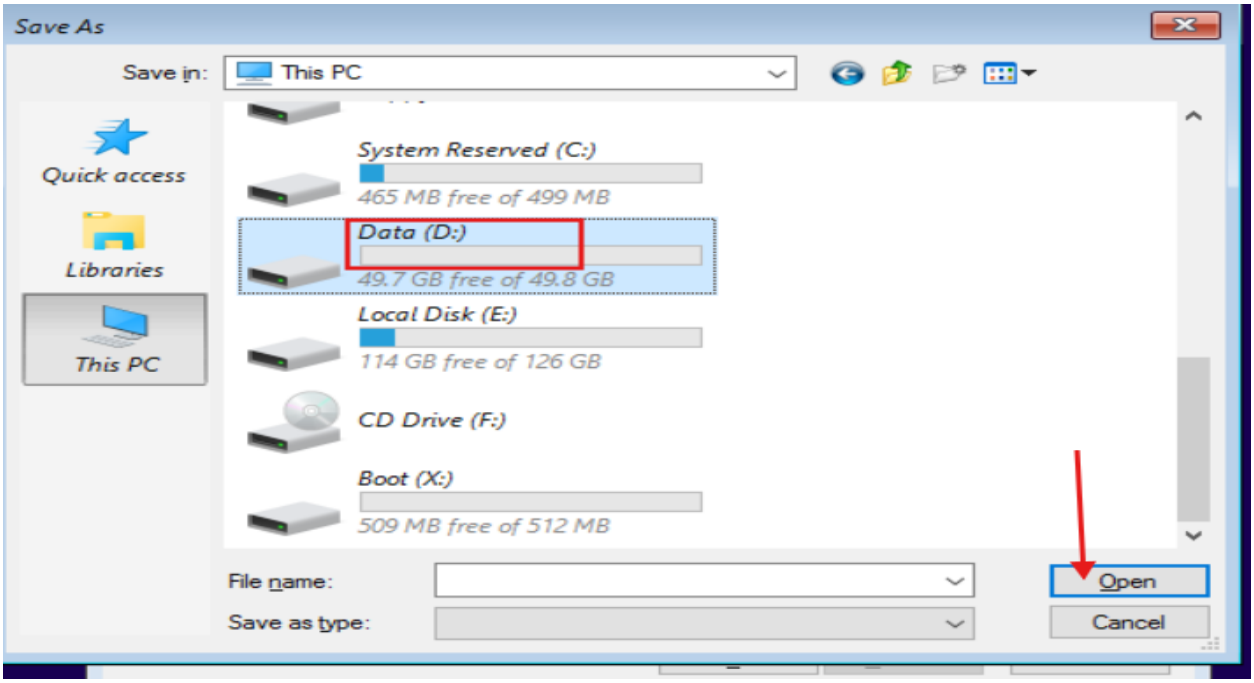
In this console, simply click the 'Next' button.



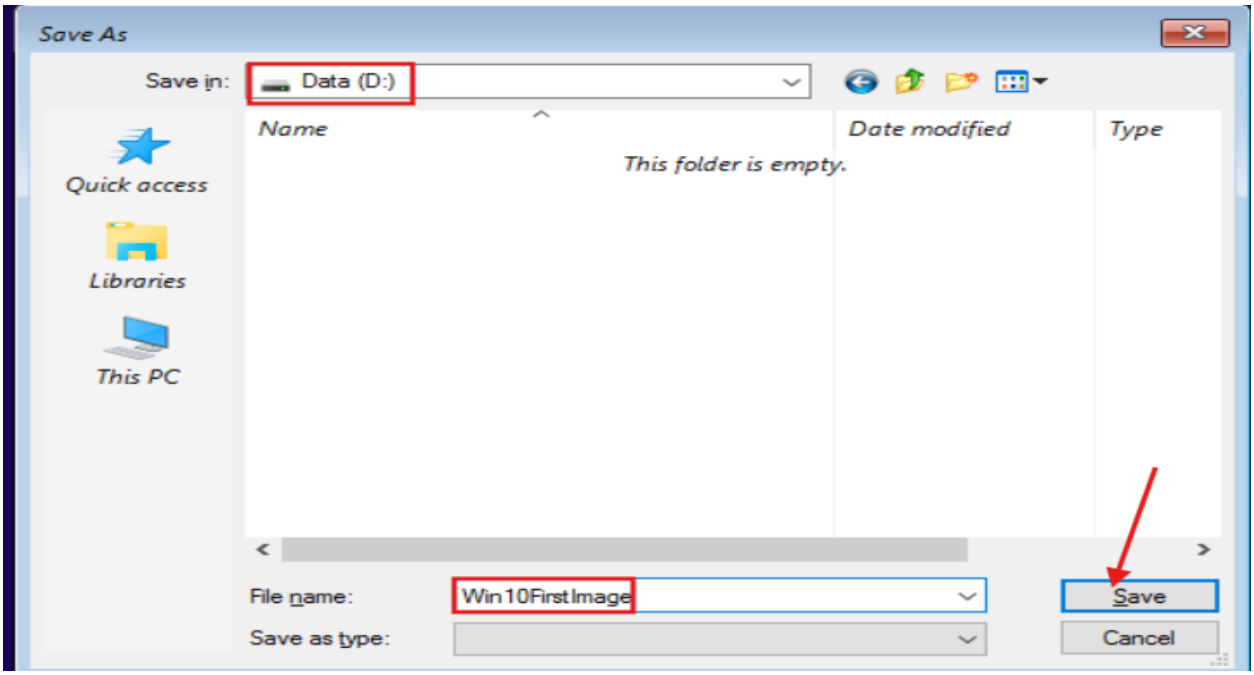
Here, it will prompt you to select the volume to capture. Open the drop-down menu and select the desired volume. Then, provide the name and description of the image, and click 'Next.'



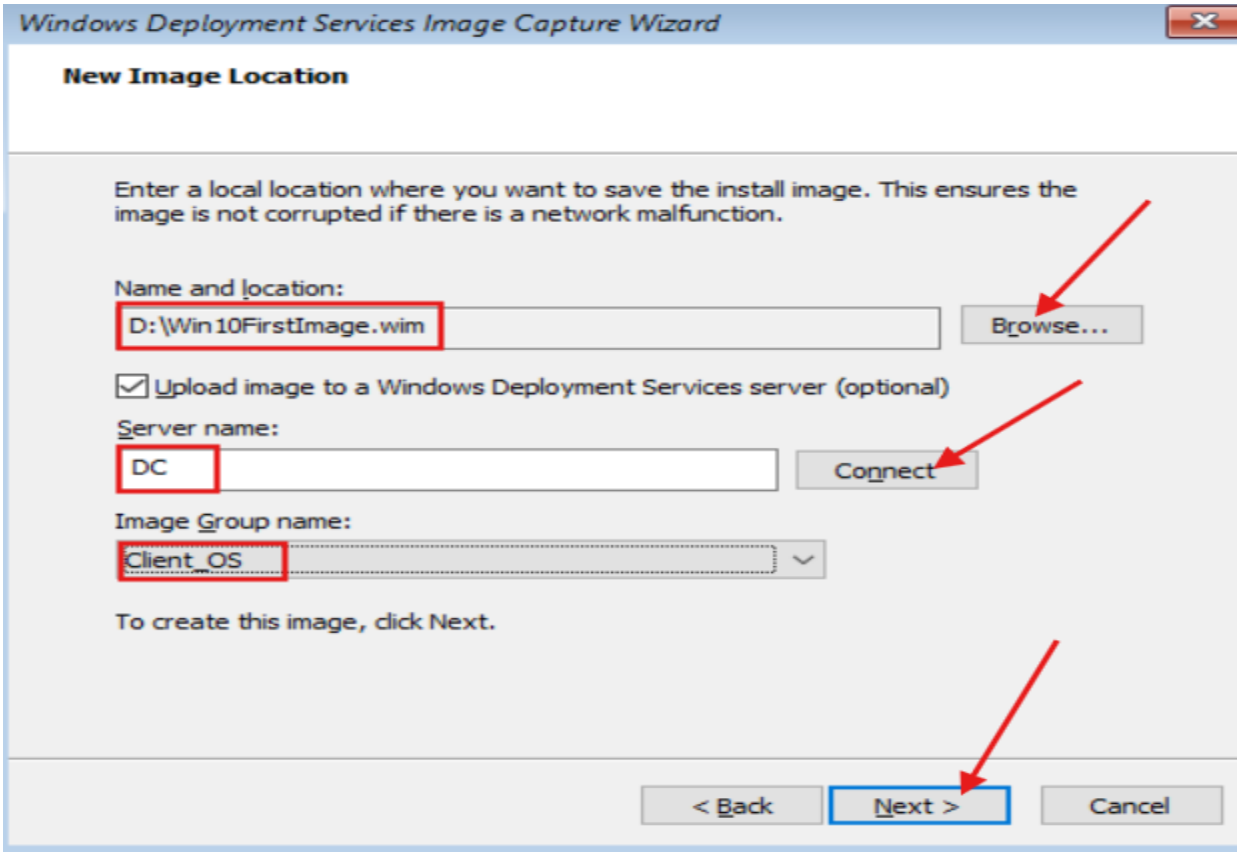
Here, it will ask for the name and location where you want to store this image, as you need to store it locally first by default.



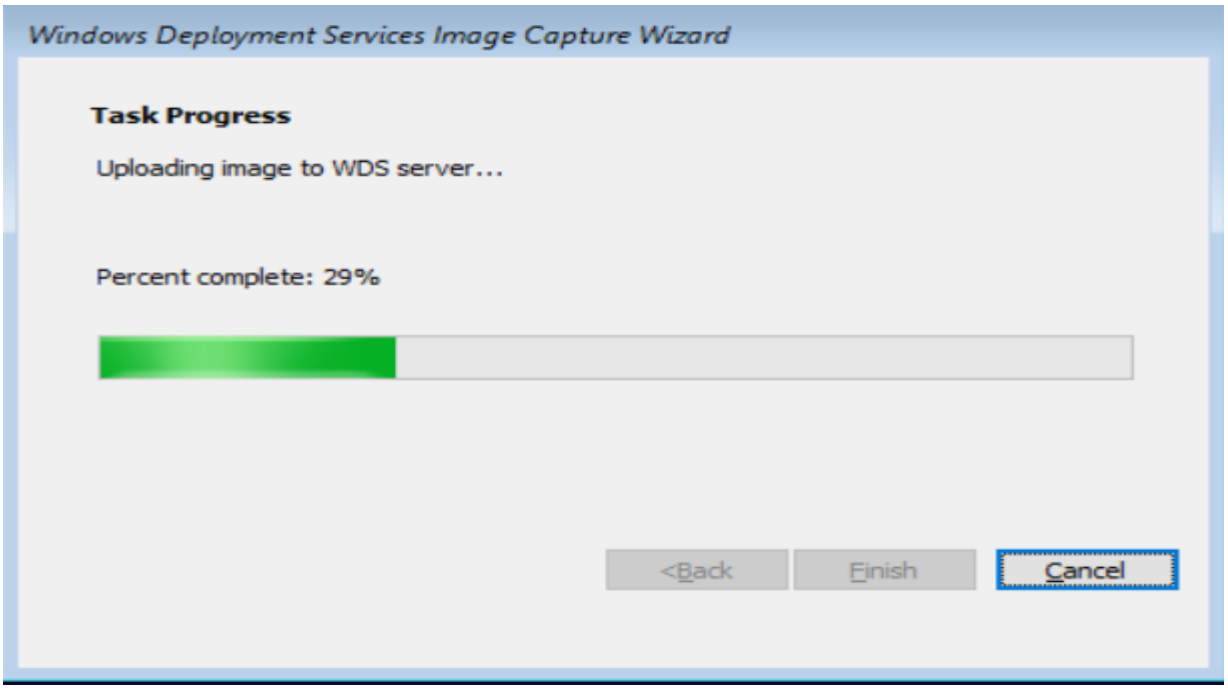
Once you select the location, you need to specify the file name.



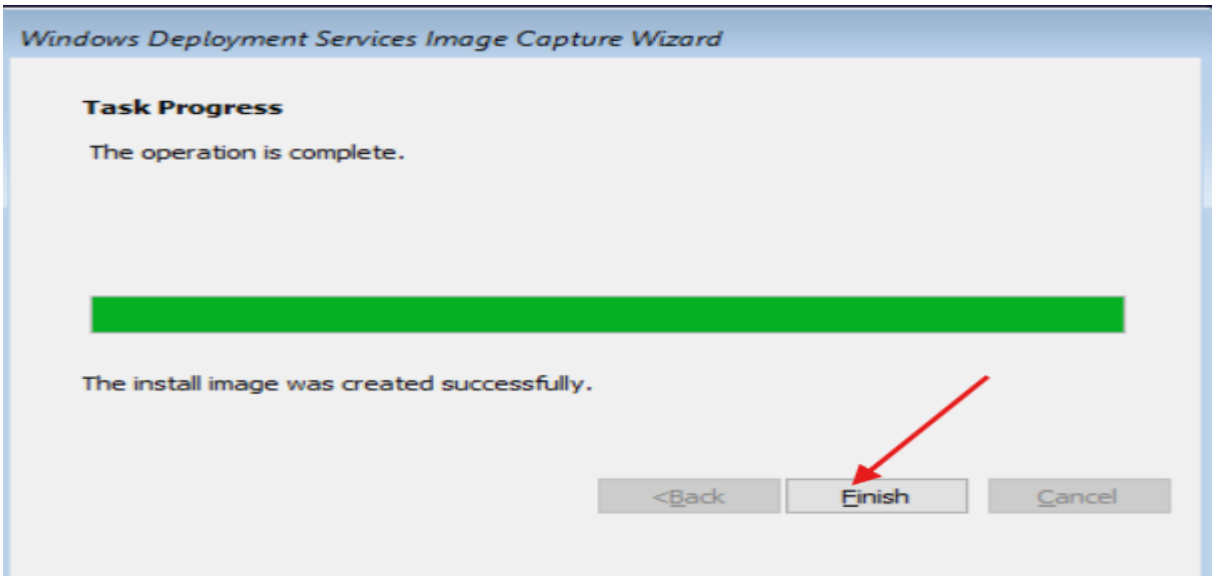
In this wizard, after specifying the location for the image, you need to connect to your server. It will prompt you for credentials. Then, select the image group name, which is 'Client_OS' and click Next.



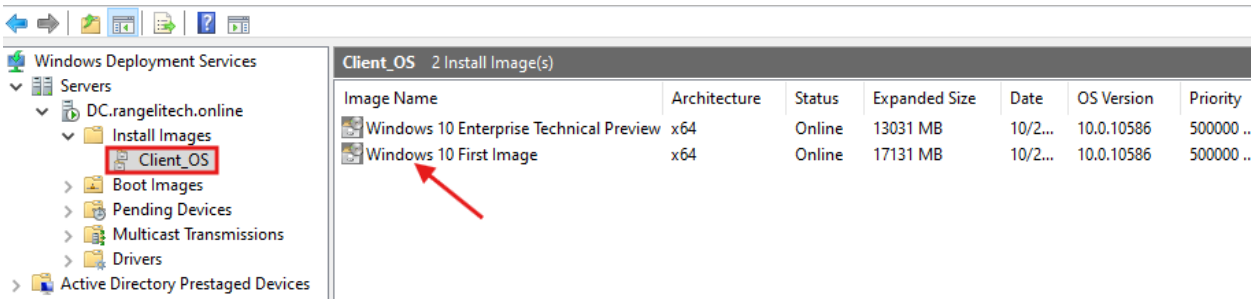
This process will take some time to complete. After that, it will begin uploading the image to the WDS server.



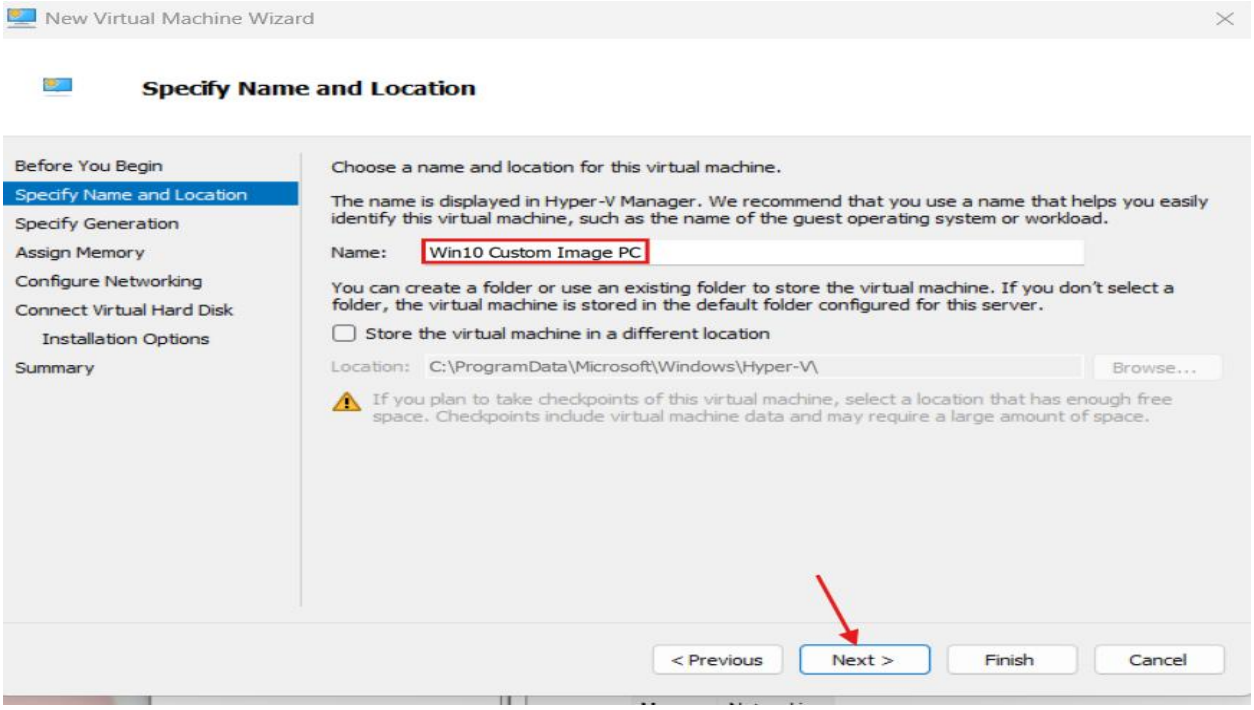
Once the operation is completed, you will see this screen. Click the 'Finish' button.



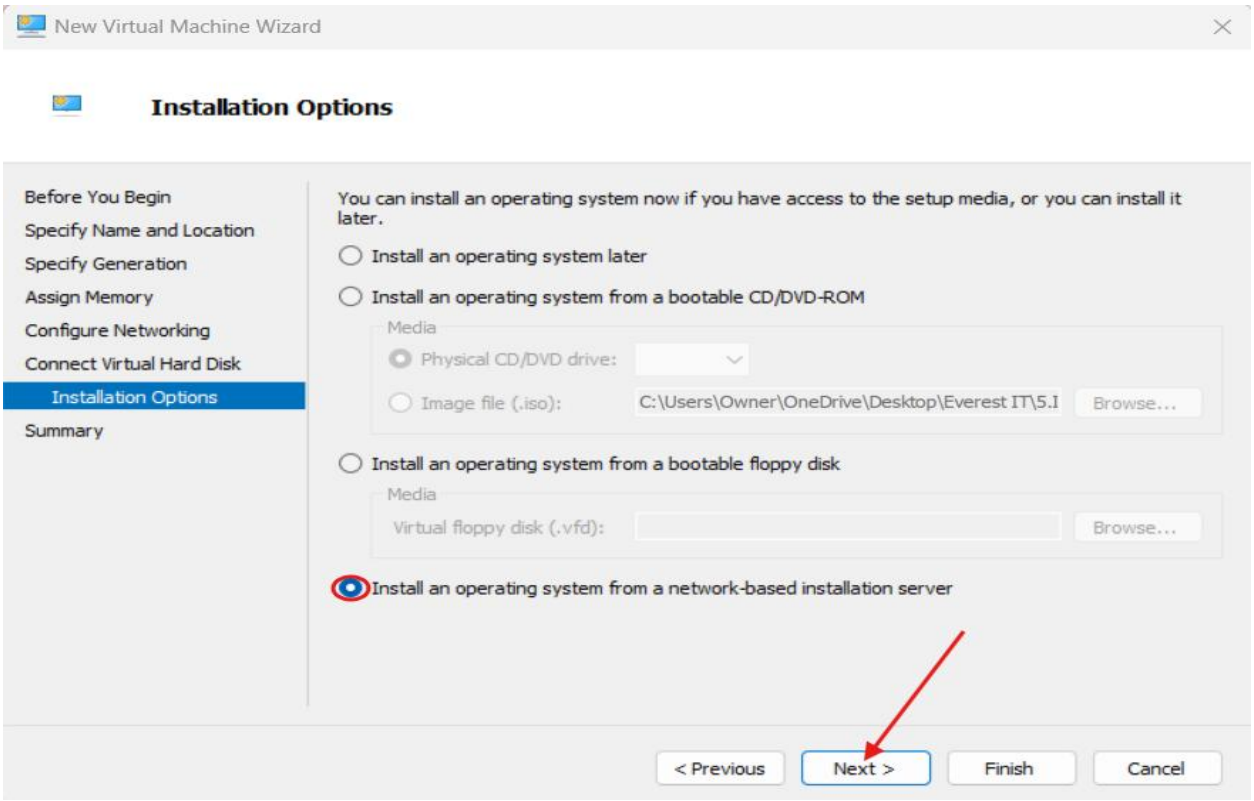
Let's verify in the WDS server. Open the WDS management console, and here you can see that the custom image has been added. This means we have successfully captured the Windows 10 reference image and uploaded it to the WDS server. Now we can use this Windows 10 image to deploy to Windows 10 client computers on the network using the WDS server, along with those custom applications.



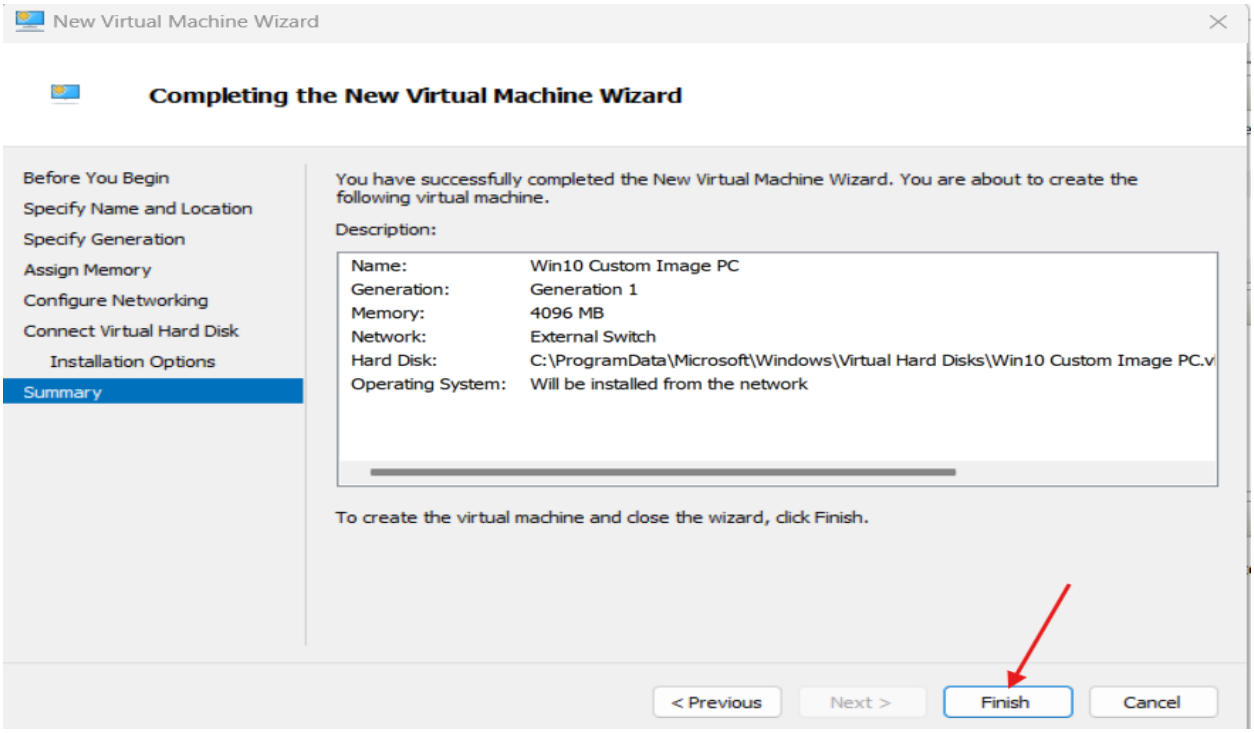
Now, go to your Hyper-V Manager, and let's deploy a new Windows 10 VM using the custom captured image.



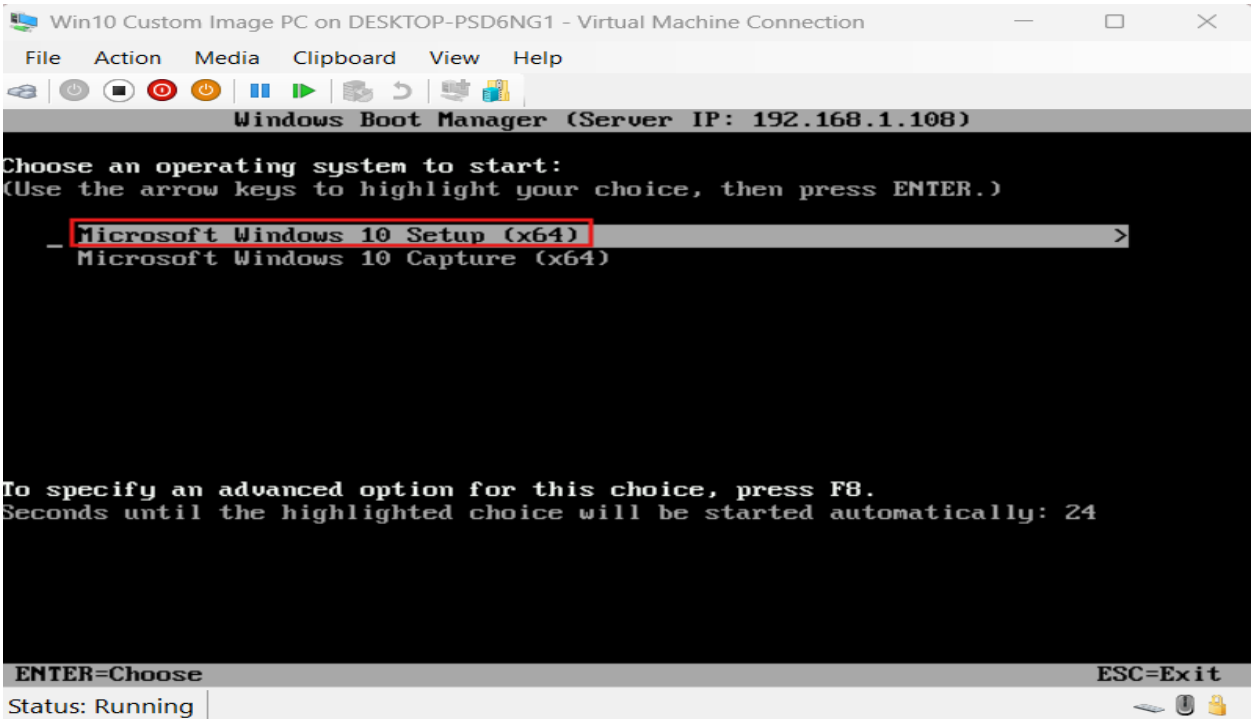
Here, in Installation Options, select Install an operating system from a network-based installation server because we want to boot this VM using the network.



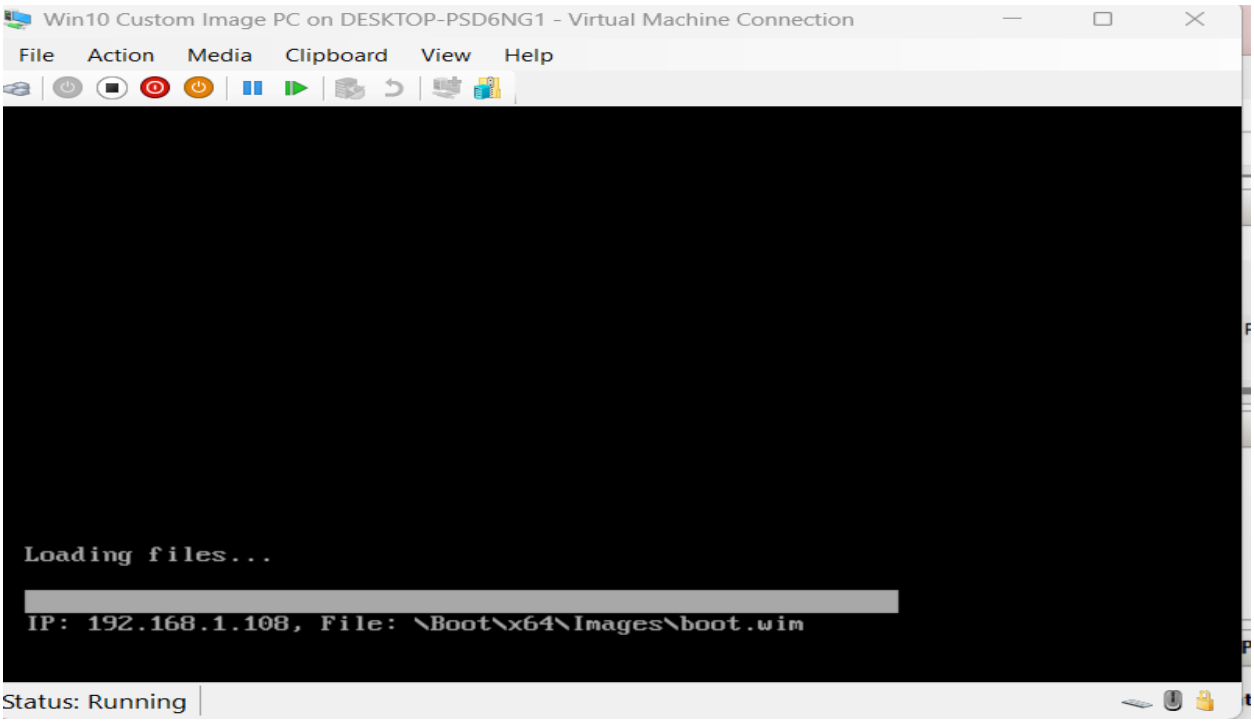
Click Finish.



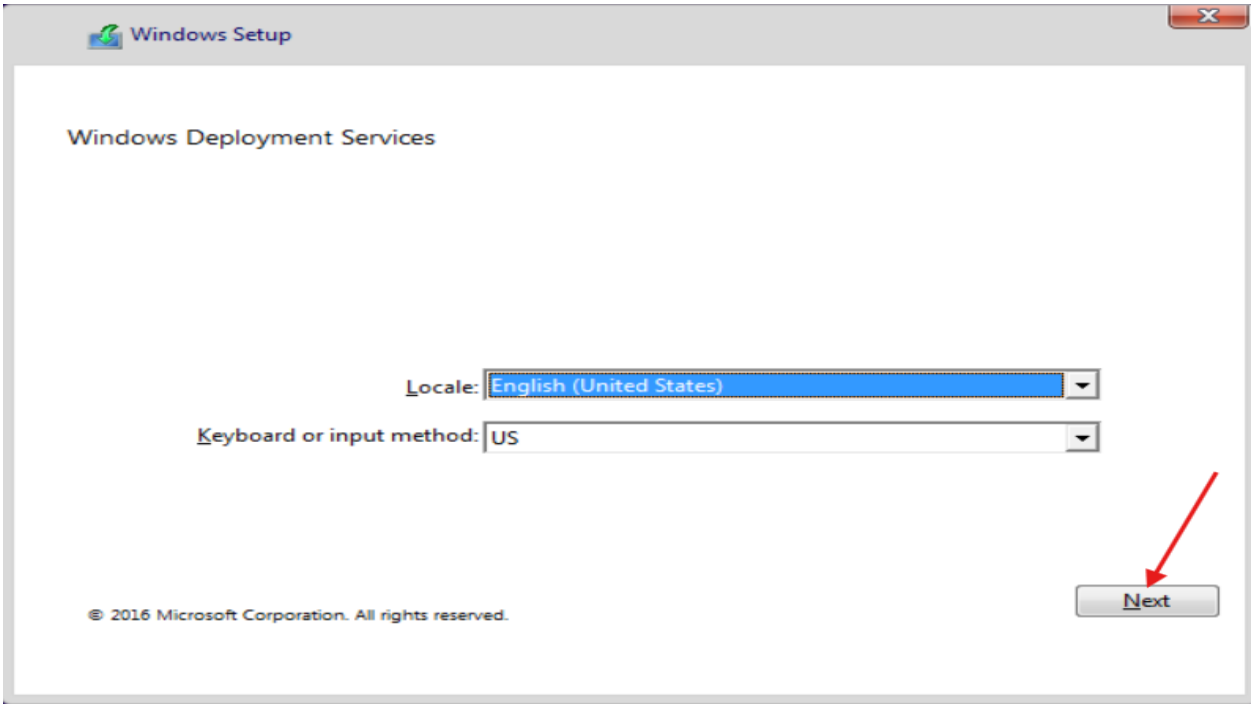
Here, you need to select the 'Microsoft Windows 10 Setup (x64)' option to install the Windows 10 operating system on the client computer.



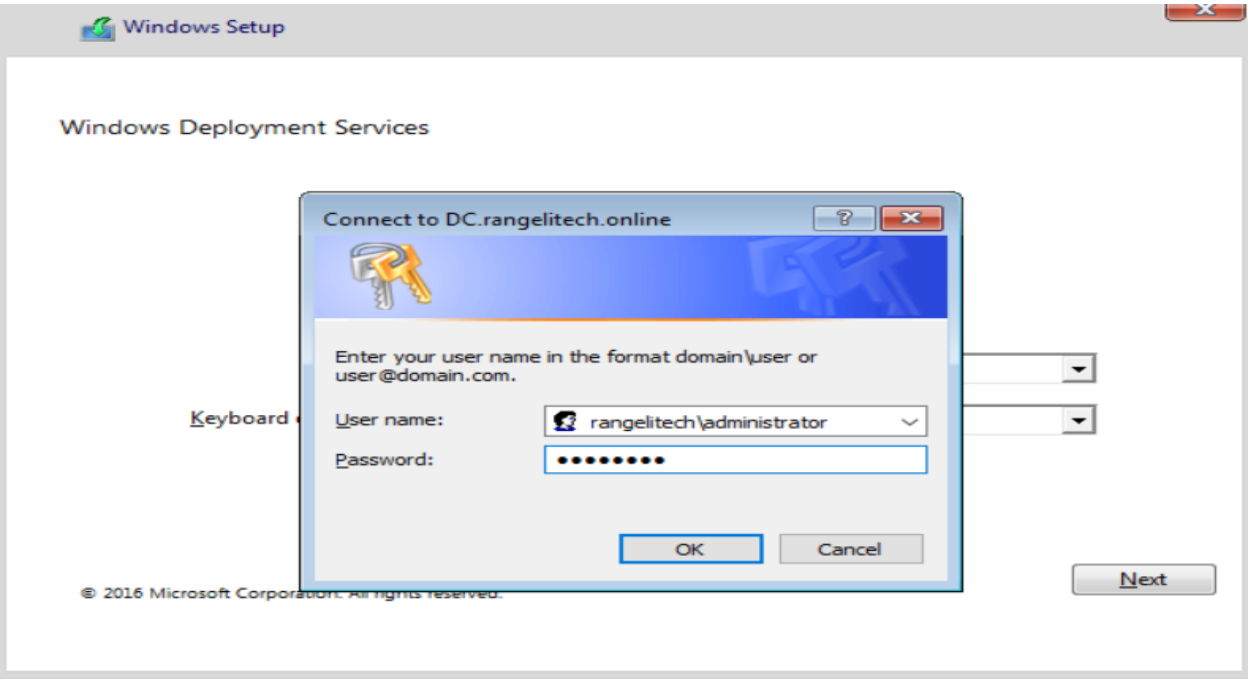
The client machine loads the boot.wim from the WDS server.



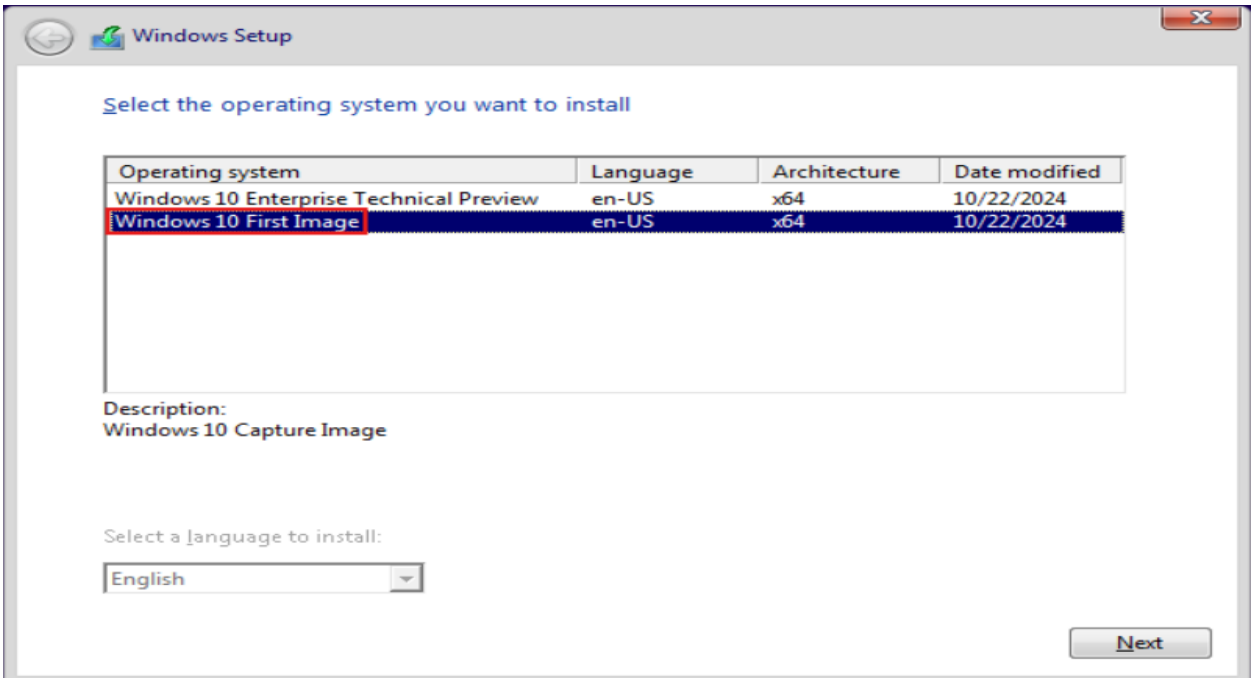
In this wizard, you can choose the applicable option and click Next.



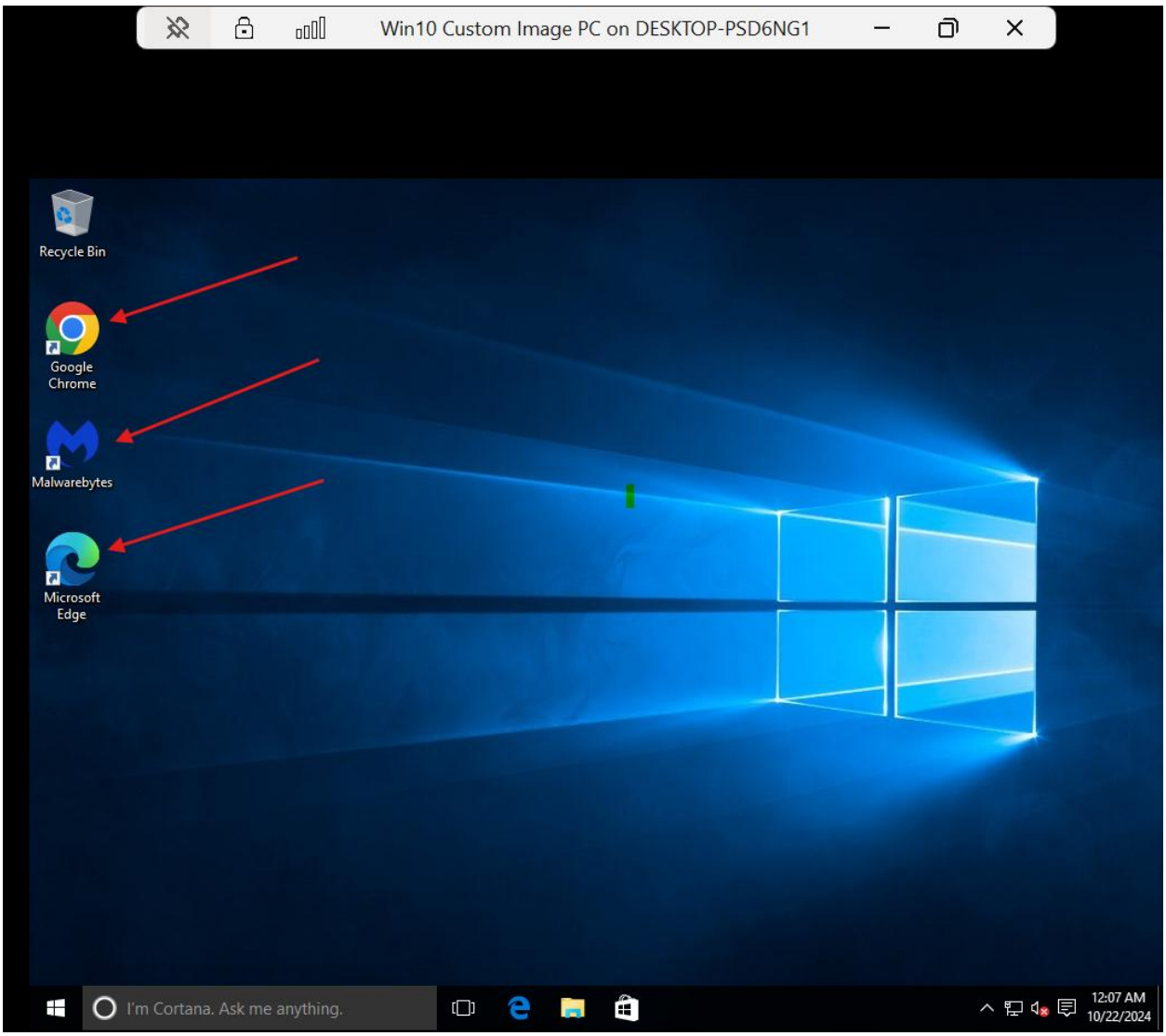
Here, you will be asked to enter the server credentials. Enter them and click OK.



In this wizard, select the 'Windows 10 First Image' option and click 'Next.' This option will install the custom captured image on the client computer.



After deploying the new VM, you can see that a new VM has been created with the Windows 10 operating system and custom loaded images.



Conclusion:

Windows Deployment Services (WDS) is essential for organizations as it streamlines the deployment of Windows operating systems across multiple devices. By enabling network-based installations, WDS reduces the need for physical media and manual setups, saving time and minimizing errors.

With WDS, administrators can create and manage custom images that include necessary applications and configurations, ensuring consistency in deployments. Additionally, its automated deployment features help maintain productivity and keep systems updated with the latest security patches. Overall, WDS enhances efficiency and simplifies IT operations, making it a vital asset for any organization.